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Navigating the gambit: Unfavorable market responses to AI-based service patents in manufacturing firms

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ABSTRACT

Drawing on transaction cost economics and organizational information processing theory, we challenge the espoused benefits of AI in servitization by demonstrating that stock market reactions to service technology patents with high AI scores are negative. In a sample of 1306 service technology patents and 758,143 non-service technology patents from 3899 manufacturing firms (1980–2020), we find that idiosyncratic volatility mitigates this negative reaction, while tangibility shows no such effect. Decade-wise analysis reveals effects are significant only in the 2010s and 2020s, suggesting evolving market perceptions of AI-driven servitization. Results hold across matched-pair sampling, non-linear effects testing, fixed-effects individual slopes, and alternative moderator analyses. This study cautions against assuming short-term market value of AI service innovations in manufacturing.

1. Introduction

The manufacturing industry is undergoing a servitization revolution, shifting focus from solely producing goods to offering a comprehensive portfolio of services alongside physical products (Baines et al., 2009; Kohtamäki et al., 2019a; Rabetino et al., 2018; Raddats et al., 2019; Vandermerwe and Rada, 1988). Artificial Intelligence (AI)-powered service technology empowers manufacturing firms to not only enhance efficiency and productivity but also embark on a path to deliver exceptional customer experiences and establish long-term value-creation strategies (Akter et al., 2023; Bustinza et al., 2018; Gebauer et al., 2021; Sjödin et al., 2023). AI service technology can help manufacturers automate repetitive tasks, streamline workflows, and optimize resource utilization, leading to tangible improvements in operational efficiency, cost reduction, and increased productivity (Kharlamov and Parry, 2021; Paschou et al., 2020). AI-powered services can handle logistics, automate maintenance scheduling, and personalize customer interactions, ultimately contributing to a seamless and efficient service delivery model. By responding quickly to inquiries and resolving issues efficiently, AI could enhance customer satisfaction and facilitate a positive experience throughout the service lifecycle.

Due to challenges in the implementation of AI in services, in general, and servitization in particular, we ask whether AI service patents lead to negative stock market reaction and whether this negative reaction can be mitigated by factors that improve organizational information processing—idiosyncratic volatility and tangibility. From an organizational information processing theory (OIPT) perspective (Galbraith, 1973; Kroh et al., 2018), the increased complexity of servitization introduced by OIPT posits that organizations have a finite capacity to process and utilize information, and excessive complexity can overload this capacity, leading to decision challenges, recombination of operational resources, decreased efficiency, and lower performance. In the context of servitization, AI-based service patents introduce an added layer of complexity that can divert resources away from core servitization activities, such as developing and delivering value-adding services and building customer relationships (Jorzik et al., 2024; Stock and Tatikonda, 2008). AI-based service patents can be attributed to the additional cognitive and logistical demands placed on manufacturing firms (Wan et al., 2020). Additionally, managing a portfolio of AI-powered service patents requires specialized expertise in managing manufacturing and service-related resources. Moreover, the ambiguity and evolving nature of AI technology can further complicate the interpretation and rebundling of operational resources.

From a transaction cost economics and organizational information processing theory (OIPT) perspective, idiosyncratic volatility (Dai, 2021; Xu and Malkiel, 2003) and tangibility (Kamp and Parry, 2017; Luiz Corrêa et al., 2007) can help mitigate the negative reactions associated with the approval of AI-based service patents. Idiosyncratic volatility, also or unsystematic risk, can help manufacturing firms maintain a focus on their core servitization activities thus reducing

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investor sensitivity to patent-related risks. When a firm's stock price is more volatile firms could focus on the long-term strategic value of their patents to develop newer initiatives that may lower long-term unsystematic risk. Tangibility is the ratio of tangible assets to total assets of the firm. In the context of AI-powered services, tangibility can provide a more concrete basis for implementing AI-based service initiatives. By leveraging the tangibility of AI-powered services, manufacturing firms can streamline manufacturing and services processes and reduce the load associated with the implementation of AI-powered services.

Despite extensive research on servitization benefits and growing attention to AI's enabling role, a critical gap remains: we lack systematic evidence on how capital markets value AI-enabled service innovations in manufacturing contexts. While operational studies emphasize efficiency gains and customer experience improvements, they implicitly assume these benefits translate into shareholder value. However, the servitization paradox literature (Brax et al., 2021; Kohtamäki et al., 2020) demonstrates that service transformation can simultaneously create and destroy value. This paradox may be amplified when AI introduces additional technological and organizational complexity. Our study addresses this gap by examining stock market reactions—a real-time aggregation of investor expectations—to AI-based service patent approvals. This market valuation lens complements operational performance research and reveals whether investors share the academic community's optimism about AI servitization.

Our study aims to make the following contributions. First, the potential impact of AI on the servitization landscape is undeniable. While AI promises to enhance service innovation, improve customer experiences, and drive value creation, its introduction also presents challenges that require careful consideration (Abou-Foul et al., 2023; Kohtamäki et al., 2020). One such challenge lies in the potential negative effects of AI service patents on servitization efforts. The counterintuitive notion challenges the conventional assumption of AI as an unequivocal catalyst for servitization success (Brax et al., 2021; Dmitrijeva et al., 2022). Our study also highlights the mitigating factors. Idiosyncratic volatility can play a mitigating role by reducing investor sensitivity to patent-related risks allowing firms to focus on long-term strategic value creation rather than being swayed by short-term market fluctuations. These insights underscore the importance of organizational information processing capabilities in shaping servitization outcomes. Manufacturing firms must effectively manage their information processing resources consider challenges in navigating the complexities of AI-driven servitization and mitigate potential negative reactions. While AI service patents may provide protection, they may also introduce uncertainties and complexities that could harm investors in the servitization context. By countering the notion that AI is an unambiguously positive force in servitization, we invite further exploration of the potential downsides of AI in servitization and the need for a balanced approach that considers both opportunities and challenges (Kohtamäki et al., 2020).

Second, the research question expands the scope of Transaction Cost Economics and Organizational Information Processing Theory (OIPT) by applying its principles to the emerging field of AI-driven servitization (Agarwal et al., 2022; Flynn and Flynn, 1999; Galbraith, 1973; Raddats et al., 2022; Stock and Tatikonda, 2008). Effective information processing strategies culminate in improved servitization outcomes, encompassing enhanced customer experiences, increased service innovation, and streamlined service delivery processes. A well-oiled information processing and operations system enables firms to gather and analyze customer feedback effectively, identify emerging market trends, and proactively adapt their service offerings to meet evolving customer needs. Furthermore, effective information processing fosters a culture of innovation, driving the development of new and improved service offerings.

2. Theoretical development hypotheses

Servitization has attracted scholarly attention across diverse disciplines, including operations management, innovation management, technology management, and marketing (Baines et al., 2009; Neely, 2008; Xing et al., 2023). Conceived as an avenue for manufacturing firms to augment their competitiveness by integrating services into their product offerings (Baines et al., 2017), servitization delineates service categories into basic, intermediate, and advanced services. Baines et al. (2009) conducted a literature review from 1976 to 2008, focusing on the concept, origin, features, and financial, strategic, and marketing drivers. Beuren et al. (2013) contributed insights into the features, benefits, and drawbacks of the product-service system through a systematic literature review from 2006 to 2010. Their findings suggest that providers of product-service systems benefit from heightened competitiveness, improved innovation potential, and increased profit. Raddats et al. (2019) conducted a systematic literature review involving 219 studies published from 2005 to 2017. Their findings indicate that services may not be profitable for manufacturers unless a critical mass is reached, while clarity is lacking on the implementation path to improved service performance.

Lexutt (2019) through a systematic literature review from 1988 to 2015 identified three groups of factors—company-related, customer-related, environment-related—that collectively influence the success of the service transition. Khanra et al. (2021) employed co-citation analysis on 275 articles spanning 2004 to 2019, concluding that further enrichment is needed in understanding the literature on the revenue and profit generation of servitized business models. Rabetino et al. (2021) utilized dynamic topic modeling on 550 articles from 2001 to 2018 and infer a substream evolving from strategies for competitive advantage in industrial services to risk management is strategically aligned for a profitable service transition in manufacturing companies. Lastly, Ulaga and Kowalkowski (2022) that service performance is contingent on various factors, including service type, development stage, and firm capabilities.

The meta-analyses provide mixed evidence. In a meta-analysis of 41 studies, Wang et al. (2018) find a positive association between servitization and performance, however, this relationship is affected by the operationalization of constructs (servitization and performance) and control variables (industry and region). A more recent meta-analysis by Faramarzi et al. (2023) encompassing 85 studies and 379 effect sizes found a small but significant servitization–performance correlation of 0.064. Exploring the influence of market type, the meta-analysis revealed that while firms across all industries generally benefit from servitization, Business-to-Business (B2B) firms derive the most significant advantages. Additionally, high-tech industries and emerging markets are identified as particularly conducive to implementing servitization strategies.

2.1. AI and service technology patents

The relationship between service technology and servitization has been a focal point of investigation in various studies. Table 1 presents the service technology patent classes (Geum et al., 2017). AI is empowering manufacturing firms to develop innovative service offerings, enhance customer experiences, and optimize operational efficiency, fundamentally reshaping the nature of service-oriented business models. AI-based servitization provides manufacturing firms with the tools and capabilities to develop new and enhanced services that meet the evolving needs of customers. AI-powered analytics and machine learning algorithms enable firms to gain deeper insights into customer behavior, preferences, and emerging trends (Gebauer et al., 2021). These insights empower firms to tailor services to specific customer segments with greater precision and personalization, fostering stronger customer relationships and enhancing brand loyalty. By automating mundane and repetitive tasks, AI enables employees to focus on higher-order activities, such as strategic planning, customer engagement, and innovation, propelling the development of cutting-edge service offerings that redefine the servitization landscape (Alexiev et al., 2018; Sjödin

Table 1
Technology classes for service technology patents.

Category	IP-4 Class	Description
Equipment-supporting technology	A47B	Tables; desks; office furniture; cabinets; drawers
	A47F	Special furniture, fittings, or accessories for shops, storehouses, bars, restaurants, or the like
	A47G	Household or table equipment
	A47J	Kitchen equipment; coffee mills; spice mills
Healthcare technology	A61B	Diagnosis; surgery; identification
Game and entertainment technology	A63F	Card, board, or roulette games; indoor games using small moving playing bodies
Machinery technology	B62B	Hand-propelled vehicles
	B65B	Machines, apparatus or devices for, or methods of, packaging articles or materials
	B65D	Containers for storage or transport of articles or materials
Building technology	E04H	Buildings or like structures for particular purpose
	E21B	Earth or rock drilling
Fluid technology	F04F	Pumping of fluid by direct contact of another fluid or by using inertia of fluid to be pumped
Measurement technology	G01C	Measuring distances, levels or bearings
	G01R	Measuring electric variables
	G05F	Systems for regulating electric or magnetic variables
Data processing technology	G06F	Electric digital data processing
	G06G	Analogue computers
	G06Q	Data processing systems or methods, specially adapted for administrative, commercial, financial, managerial, supervisory or forecasting purposes
	G06K	Recognition of data; presentation of data
	G08B	Signaling or calling systems
Financial and trading technology	G07F	Coin-freed or like apparatus
	G07G	Registering the receipt of cash, valuables, or tokens
Educational technology	G09B	Educational or demonstration appliances
Communication technology	H01R	Electrically conductive connections
	H04M	Telephonic communication
	H04Q	Selecting
	H04L	Transmission of digital information
	H04B	Transmission
	H04N	Pictorial communication
	H04W	Wireless communication networks
	H04J	Multiplex communication
	H04H	Broadcast communication
	H04R	Loudspeakers, microphones, gramophone pickups
	H05B	Electric heating

et al., 2023). AI is revolutionizing the customer experience in the servitization era by providing personalized, proactive, and predictive support, transforming customer interactions, and fostering long-term customer satisfaction. AI-powered chatbots and virtual assistants serve as intelligent customer support agents, handling inquiries, resolving issues, and providing recommendations in a timely and efficient manner, enhancing customer satisfaction and loyalty.

AI is not only transforming the customer experience but also revolutionizing operational efficiency within manufacturing firms, leading to streamlined processes, reduced costs, and improved profitability (Raddats et al., 2019). AI-powered algorithms can identify and resolve operational bottlenecks, optimize resource allocation, and predict demand patterns, leading to significant efficiency gains and cost savings (Kinkel et al., 2022). Furthermore, AI can optimize inventory management, supply chain operations, and logistics, further enhancing operational efficiency and reducing costs. By automating inventory replenishment, optimizing transportation routes, and streamlining logistics processes, AI enables firms to operate with greater efficiency and responsiveness, gaining a competitive edge in the servitization landscape.

We now turn to developing our hypotheses. Table 2 provides a review of each hypothesis through the lenses of TCE and OIPT.

2.1.1. Patent signals and investor inference

A fundamental question for understanding market reactions to AI service patents concerns how investors infer organizational strain from patent grants when patents represent potential capabilities rather than realized implementations. While patents do not guarantee deployment, they signal strategic intentions and resource commitments that capital markets must evaluate. We develop our theoretical framework by examining five mechanisms through which patent approvals convey information about anticipated organizational challenges.

Patent filing and prosecution require substantial R&D investment already incurred by the time of patent approval. The United States Patent and Trademark Office estimates average costs of $10,000–$15,000 for utility patents when including attorney fees, filing fees, and prosecution costs, with complex patents often exceeding $30,000. Patent approval thus confirms that the firm has committed meaningful resources to AI service development, triggering investor assessments of opportunity costs, future implementation capital requirements, and potential diversion from core manufacturing competencies. Unlike R&D announcements that may represent exploratory research with minimal commitment, granted patents signal that ideas have progressed through formal intellectual property protection, suggesting more concrete development intentions. Investors rationally interpret this escalation of commitment as increasing the likelihood of subsequent implementation investments that may strain organizational capabilities.

Patent claims explicitly describe technological requirements, enabling investors to directly assess gaps between the firm's current organizational capabilities and the patent's specifications. When manufacturing firms patent AI-driven predictive maintenance services requiring machine learning model training and deployment, natural language processing for customer interfaces, or recommendation algorithms for service personalization, these technical requirements become publicly observable through patent disclosures. Manufacturing firms typically lack several capabilities essential for AI service delivery: data science teams with expertise in statistical learning and algorithm development, cloud computing infrastructure for scalable AI model deployment, software engineering capabilities for building service applications, and customer success organizations for ongoing service relationships. Patent disclosures make these capability deficits salient to investors, who can compare patent specifications against known organizational capabilities. The technical specificity of patent claims thus serves an unintended information revelation function, allowing investors to gauge implementation challenges with greater precision than general strategic announcements would permit.

AI service patents introduce dual uncertainties that distinguish them from traditional product patents. For product patents in manufacturing firms' core domains, investors can reasonably assume the firm possesses established production capabilities, supply chain relationships, and distribution channels necessary for commercialization. Service patents eliminate this assumption, creating two distinct sources of uncertainty. First, technological uncertainty concerns whether the firm can reliably deploy AI at scale: Can data pipelines be established? Will models maintain accuracy across diverse customer environments? Can the firm manage continuous model updating and retraining? Second, commercial uncertainty concerns customer adoption and monetization: Will customers trust AI-generated recommendations? What price will the market bear for AI services? How will services be packaged and delivered? Patent approval resolves neither uncertainty. Indeed, patent grants may heighten uncertainty by confirming the firm's serious intent to enter uncertain territory, triggering investor concerns about capital

Table 2
Conceptual rationale.

	Theory	Rationale
Hypothesis 1. Approval of service technology patents scoring high on AI will be associated with a negative stock market reaction.	Transaction Cost Economics (TCE)	Increased uncertainty and complexity: AI-related service technology patents can introduce new levels of uncertainty and complexity due to the evolving nature of AI technology and its potential impact on the firm's business model. Investors may perceive this heightened uncertainty as a risk factor, leading to a negative market reaction. Higher safeguarding costs: The development and implementation of AI-based service technologies may necessitate additional safeguarding measures to protect intellectual property, prevent data breaches, and ensure compliance with regulatory requirements. These increased safeguarding costs can raise transaction costs and potentially negatively impact investor sentiment. Opportunism concerns: AI-related partnerships often involve collaborations with external AI providers, introducing potential opportunism risks. Investors may be wary of these risks, as opportunism can lead to value misappropriation and reduced returns on AI investments.
	Organizational Information Processing Theory (OIPT)	Organizational disruption: AI-based service technologies can disrupt established organizational routines, processes, and decision-making structures, requiring significant adaptation and change management efforts. This disruption can strain organizational cognitive capacity and overwhelm information flows, leading to temporary inefficiencies and performance fluctuations. Cognitive salience and familiarity: AI-based service technologies often involve intangible concepts and complex algorithms, which may hinder their cognitive salience and familiarity among organizational decision-makers. This lack of understanding can lead to misinterpretations, errors, and delayed responses, further exacerbating the challenges associated with AI integration. Information overload: The adoption of AI-based service technologies can generate vast amounts of data and information, potentially overwhelming existing channels for information flows and sensemaking. This information overload can strain the organization's ability to process and interpret relevant data, leading to suboptimal decision-making.
Hypothesis 2. For firms with higher idiosyncratic volatility, approval of service technology patent scoring high on AI will be associated with a less negative stock market reaction.	Transaction Cost Economics (TCE)	AI adoption introduces increased uncertainty and complexity, potentially leading to higher transaction costs that negatively impact investor sentiment. Idiosyncratic volatility: Firms operating in highly volatile environments may already face significant uncertainty due to external factors, potentially mitigating the negative market reaction to AI adoption. Investors may already account for these risks in their valuation of the firm, reducing the additional impact of AI-related uncertainty.
	Organizational Information Processing Theory (OIPT)	AI adoption can disrupt established organizational routines and processes, straining cognitive capacity and overwhelming information flows, potentially leading to temporary inefficiencies and performance fluctuations that negatively impact investor perceptions. Idiosyncratic volatility: Firms with a history of operating in volatile environments may have developed stronger organizational capabilities for handling uncertainty and change. These firms may be better equipped to adapt to AI-induced disruptions, potentially reducing the negative market reaction.
Hypothesis 3. For firms with higher tangibility, approval of service technology patents scoring high on AI will be associated with a less negative stock market reaction.	Transaction Cost Economics (TCE)	Tangible assets associated with AI adoption, such as specialized hardware or dedicated IT infrastructure, can facilitate easier monitoring and performance verification, provide collateral for AI partnerships, and simplify valuation for investors, potentially reducing transaction costs and mitigating investor concerns.
	Organizational Information Processing Theory (OIPT)	Tangible assets associated with AI adoption can enhance cognitive salience, proximity, and familiarity, potentially facilitating better understanding, integration, and acceptance of AI within the organization.

allocation to risky ventures outside the firm's established competencies.

Many AI service patents describe systems requiring integration across multiple organizational functions and external boundaries. Predictive maintenance services, for example, require coordination among IT infrastructure teams deploying cloud services, manufacturing operations sharing equipment data, customer service managing client relationships, sales teams communicating value propositions, and potentially external partners providing specialized AI capabilities. Patent claims often explicitly detail these integration requirements through system diagrams, data flow specifications, and interface descriptions. Manufacturing firms with traditionally siloed functional organizations may lack the horizontal coordination mechanisms necessary for such cross-functional integration. Investors familiar with organizational theory recognize that coordination costs increase exponentially with the number of interdependent units, creating skepticism about whether hierarchical manufacturing organizations structured for sequential production processes can manage the lateral coordination AI services demand. The patent claims themselves, particularly in complex systems patents with numerous dependent claims and integration specifications, reveal coordination requirements that investors may doubt the organization can manage effectively.

Not all patents carry equal weight as implementation indicators, and investors likely discriminate based on patent characteristics. Broad patents with generic claims may signal defensive intellectual property strategies aimed at blocking competitors or establishing negotiating positions in patent portfolios, with minimal actual deployment intent. Such patents impose limited organizational strain because the firm may never attempt implementation. Conversely, patents exhibiting several characteristics more credibly indicate serious implementation plans: detailed technical specifications describing specific algorithms, architectures, or protocols; narrow, use-case-specific claims suggesting concrete application scenarios; substantial prior art citations demonstrating deep engagement with existing technologies; and continuation applications or patent families indicating sustained development efforts. These characteristics transform the patent from a defensive chess piece into a credible signal of implementation intent, thereby amplifying investor concerns about organizational strain. Ironically, higher-quality patents that more convincingly demonstrate technical merit may generate more negative market reactions precisely because they more credibly forecast challenging implementations.

From an investor perspective, the patent itself becomes information about future organizational challenges even absent explicit confirmation of implementation plans. The information asymmetry between firm management and external investors means that patent disclosures provide rare, detailed glimpses into strategic directions and technical commitments. Unlike carefully crafted earnings calls or investor presentations where management controls the narrative, patent applications undergo adversarial examination by patent examiners, requiring technical specificity that reveals implementation requirements. Investors can thus treat patents as relatively credible signals because the patent system's requirements for novelty, non-obviousness, and enablement force detailed technical disclosure. This disclosure, while intended to protect intellectual property, simultaneously reveals to investors the magnitude of organizational transformation required to commercialize the invention—transformation that, as we develop in subsequent sections, may exceed manufacturing firms' adaptive capacity and create persistent misalignment.

2.2. Service AI patents and market reaction

OIPT emerged out of Carnegie School insights into bounded rationality amidst uncertainty and decision-making constraints (Galbraith, 1973). It posits that organizations require sufficient information processing capacity to execute activities demanding cognitive attention, interpretation, and meaningful responses. Processing needs intensify under complexity from environmental turbulence or internal interdependencies (Daft and Lengel, 1986). By sculpting structures, systems, and processes around information flows, organizations resolve uncertainty to enable decisions and actions (Galbraith, 1977).

Several key concepts underpin OIPT with direct applicability to service contexts. These include conceptualizing organizations as open information processing systems (Kmetz, 2018; Kroh et al., 2018) delineating requirements for managing equivocality through sensemaking versus uncertainty via decision support (Daft and Macintosh, 1981); specifying analytical and intuitive processing modes (Burton and Obel, 2018); and recognizing loose coupling between formal and informal flows (Orton and Weick, 1990). OIPT explanations also now integrate behavioral limitations in managerial attention (Ocasio, 2001), network governance arrangements (Joseph and Gaba, 2020), and multi-level processing tensions between exploration and exploitation (Rojas-Córdova et al., 2023). The widespread expanding scope bolsters OIPT's applicability to multifaceted service implementations.

By foregrounding information requirements, OIPT bears direct relevance for service innovation and delivery spanning elevated interdependence, customization pressures, and knowledge intensity. First, greater reciprocal interdependencies in services necessitate processing mechanisms enabling coordination across units (Dalenogare et al., 2023). Through extensions like service-dominant logic emphasizing cocreation and value-in-use (Vargo and Lusch, 2008), OIPT underscores service ecosystems' demands for processing capacity across partners to minimize equivocality, absorb new knowledge, and reconcile frames (Daft and Macintosh, 1981). By spotlighting multi-level tensions, OIPT also cautions against hierarchical controls in services stifling frontline improvisation and tailoring (Vieira Da Cunha et al., 2003).

OIPT emphasizes the importance of collaboration and knowledge sharing within organizations as a means of enhancing innovation, adapting to change, and improving organizational performance (Christensen, 2018). However, the defensive nature of AI-based service patents can create barriers to collaboration and knowledge sharing among manufacturing firms. When firms hold valuable AI-powered service patents, they may be hesitant to share their knowledge and expertise with competitors for fear of losing their competitive advantage (Johnson et al., 2021). This reluctance to collaborate can hinder the development and adoption of innovative servitization solutions, as firms may be unwilling to share ideas and resources that could potentially benefit their competitors. Furthermore, the threat of patent infringement lawsuits can discourage firms from engaging in open collaboration, as they may fear legal consequences if their partners inadvertently infringe upon their patented technologies (Lim, 2018). This fear of litigation can create a culture of secrecy and distrust, further hindering knowledge exchange and slowing down the pace of servitization innovation.

While the incorporation of AI in servitization offers transformative potential, it concurrently presents a set of challenges that demand attention and realignment of resources. OITP highlights the challenge of information overload from combining AI into services may be especially challenging for manufacturing firms focused on the transformation of tangible inputs. With AI service patents generating newer manufacturing and service-related challenges, there is a potential contribution to this overload, making it challenging for organizations to effectively extract valuable insights and make well-informed decisions. The intrinsic complexity of AI technologies poses a significant hurdle, necessitating specialized expertise for the effective implementation, maintenance, and optimization of AI solutions. Firms may encounter difficulties in acquiring and retaining the requisite skills and knowledge to fully harness the potential of AI, potentially impeding its seamless adoption. Furthermore, the extensive collection and utilization of customer data for AI-powered analytics raise apprehensions regarding data security and privacy (Kohtamäki et al., 2019a). Establishing robust data security protocols, ensuring compliance with data privacy regulations, and maintaining transparency with customers becomes imperative for safeguarding sensitive information and upholding trust.

Insights from the challenges associated with AI adoption in servitization (Abou-Foul et al., 2023) contribute to several arguments supporting this perspective. TCE contends that servitization, especially involving advanced and customized services, results in higher transaction costs due to asset specificity and idiosyncratic investments. The introduction of AI service patents might exacerbate these costs. AI technologies demand specialized expertise for effective implementation and maintenance, potentially resulting in increased transaction costs related to the acquisition and retention of AI-related skills. Furthermore, TCE argues that higher asset specificity in servitization increases the risk of opportunistic behavior by buyers, leading to uncertainty and elevated transaction costs (Kohtamäki et al., 2019b; Ruiz-Martín and Díaz-Garrido, 2021). AI service patents, being specific and potentially proprietary, may expose firms to the risk of opportunistic exploitation by competitors or other market players, contributing to increased transaction costs. Moreover, TCE emphasizes the role of governance mechanisms in mitigating transaction risks (Sjödin et al., 2019). The introduction of AI service patents may necessitate the development and utilization of new governance mechanisms to protect intellectual property and manage relationships with customers and partners.

We acknowledge an important theoretical equivocality: AI capabilities could plausibly generate opposite investor expectations. AI signals automation, efficiency, and competitive advantage—factors that might drive positive market reactions, as observed when firms announce AI adoption for internal operations (e.g., AI-driven quality control or supply chain optimization). Yet our TCE/OIPT framework emphasizes complexity, transaction costs, and information processing strain—suggesting negative reactions. Which expectation dominates depends critically on context. In the AI service patent context, we theorize negative expectations prevail for three reasons. First, related to strategic alignment manufacturing firms' core competencies—production engineering, quality control, logistics—are largely orthogonal to AI service delivery requirements (data science, algorithm operations, customer relationship management, service orchestration). AI service patents may signal strategic drift rather than capability extension. By contrast, when manufacturers patent AI-driven production optimization, the alignment with existing competencies is clearer and potentially value-creating. Second, monetization uncertainty related to efficiency gains from internal AI adoption (e.g., predictive maintenance for owned factories) directly reduce costs with measurable impact. AI service patents, however, imply external customer-facing offerings where willingness-to-pay is uncertain and competitive dynamics unclear. Will industrial customers pay premiums for AI-enabled services, or will competition force price reductions? This commercialization risk is largely absent when AI improves internal operations. Third, learned skepticism our decade-wise analysis (Table 9) shows negative reactions emerged and strengthened precisely when AI hype cycles and implementation disappointments became common knowledge (post-2010). Markets may have developed domain-specific skepticism toward AI service promises while remaining optimistic about AI operational applications. The equivocality resolves differently across application domains and time periods, with service contexts facing heightened scrutiny."

In summary, the incorporation of AI service patents into servitization, particularly from TCE and OIPT perspectives, may lead to negative market reactions. From an investor perspective, AI-based service patents signal a strategic shift that activates both TCE and OIPT concerns simultaneously. Transaction cost economics suggests investors will anticipate higher monitoring costs, asset specificity risks, and governance challenges as firms navigate AI vendor relationships and complex customer contracts (Williamson, 1985). Organizational information processing theory indicates investors recognize that manufacturing firms—optimized for tangible production workflows—may lack the cognitive bandwidth to effectively integrate AI service complexities involving data management, algorithm deployment, and service delivery orchestration (Galbraith, 1973). Critically, patent approval provides limited information about implementation capabilities, realistic timelines, or customer demand validation. Unlike product patents where manufacturing expertise is well-established, service patents in AI domains require new organizational competencies (data science, machine learning operations, service relationship management) that investors may doubt manufacturers possess. The capability uncertainty manifests as negative abnormal returns at patent grant announcements.

Hypothesis 1. Approval of a service technology patent scoring high on AI will be.

associated with a negative stock market reaction.

2.3. *Mitigating effects of idiosyncratic volatility and tangibility*

Idiosyncratic volatility, also known as unsystematic risk, can play a mitigating role in reducing the negative effects of AI-based service patents on servitization for manufacturing firms (Jacobs et al., 2016; Josephson et al., 2016; Mazzucato and Tancioni, 2008). Idiosyncratic volatility refers to the volatility or fluctuation in a firm's stock price that is specific to the firm itself, rather than being driven by broader market factors (Mazzucato and Tancioni, 2008; Miller and Chen, 2003). Diversification through AI in services could reduce the impact of firm-specific events on the firm's stock price, while effective risk management strategies enable the identification and mitigation of potential risks.

From an OIPT perspective, idiosyncratic volatility can help manufacturing firms maintain a focus on their core servitization activities by reducing investor sensitivity to patent-related risks. When a firm's stock price is less volatile and more closely aligned with its fundamental business performance, investors tend to be less reactive to news or events related to AI-based service patents. This reduced sensitivity can allow firms to focus on the long-term strategic value of their patents without being overly pressured by short-term market fluctuations. In essence, idiosyncratic volatility acts as a buffer, shielding manufacturing firms from the potential distractions and resource drains associated with AI-based service patents. By fostering a more stable investment environment, idiosyncratic volatility allows firms to dedicate their cognitive resources toward the strategic implementation of servitization strategies, thereby enhancing their ability to navigate the complexities of AI-based service patents and achieve success in the servitization era.

Tangibility can play a mitigating role in reducing the negative effects of AI-based service patents on servitization for manufacturing firms (Patel et al., 2019; Villalonga, 2004). From an OIPT perspective, tangibility can help manufacturing firms maintain a focus on their core servitization activities by providing a clearer framework for intellectual property protection. When AI-powered services have a stronger tangible component, firms can more easily identify and protect their proprietary technologies, reducing the overall cognitive load associated with patent management (Alexiev et al., 2018). In essence, tangibility acts as a touchstone, providing manufacturing firms with a more tangible basis for understanding and protecting their intellectual property in the context of AI-based services. By offering a clearer framework for ownership and infringement assessment, tangibility can help firms navigate the complexities of AI-based service patents and focus their cognitive resources on the strategic implementation of servitization strategies, thereby enhancing their ability to achieve success in the servitization era.

2.3.1. *Idiosyncratic volatility and AI service technology patents*

In the realm of organizational dynamics and stock market responses, the hypothesis posits that idiosyncratic volatility mitigates the negative relationship between AI service patent approval and stock market reaction. Idiosyncratic volatility, defined as the unique risk attributed to a particular firm's stock beyond broader market movements, serves as a crucial parameter reflecting firm-specific uncertainty. Examining this hypothesis through the lens of Organizational Information-Processing Theory (OITP), a theoretical framework focused on how organizations acquire, process, and utilize information for decision-making (Daft and Lengel, 1986), reveals several interrelated aspects.

OITP underscores that the strategic utilization of insights derived from information processing significantly influences decision quality (Haußmann et al., 2012; Wickens and Carswell, 2021). Firms confronting higher idiosyncratic volatility can strategically leverage AI-generated insights from service technology patents to address specific uncertainties. As information processing becomes more strategic, it positively impacts decision-making outcomes, aligning with the OITP perspective on the role of information utilization in organizational success (Burton and Obel, 2018). OITP emphasizes the role of trust in information systems (Daft and Lengel, 1986). AI-powered service technology patents, when proven to deliver reliable and valuable insights, can foster trust in the information processing capabilities of a firm.

In conclusion, from the OITP perspective, the approval of service technology patents featuring high AI scores is posited to have mitigating effects on the negative stock market reactions associated with higher idiosyncratic volatility. Effective information processing, managing information overload, adapting information systems, strategic utilization of AI insights, and fostering trust and reliability, alongside addressing AI-related challenges, collectively contribute to the hypothesis, aligning with the principles outlined by OITP and supported by published studies.

Hypothesis 2. For firms with higher idiosyncratic volatility, approval of a service.

technology patent scoring high on AI will be associated with a less negative stock market reaction.

2.3.2. Tangibility and AI service technology patents

Hypothesis 3 posits a relationship between the approval of service technology patents, specifically those rich in artificial intelligence (AI), and the stock market reactions of firms characterized by higher tangibility. Tangibility in this context refers to the proportion of a firm's total assets represented by tangible assets, such as physical properties and equipment. Tangible assets, being physically observable, provide a degree of visibility that can influence information processing (Daniel and Titman, 2006; Wickens and Carswell, 2021). In the context of AI-rich service technology patents, tangible assets may serve as a foundational element for comprehending the potential impact and utilization of these technologies (Jancenelle, 2021). Investors may find it easier to assimilate information related to tangible assets, leading to a potentially less negative stock market reaction (Almeida and Campello, 2007; Atasoy et al., 2022).

Firms with higher tangibility may find it comparatively more straightforward to adapt their information systems to accommodate AI-rich service technology patents. Tangible assets, being physical and often associated with clear operational processes, can act as a stable foundation for integrating advanced technologies (Alexiev et al., 2018). This adaptability may contribute to a more seamless assimilation of AI technologies, potentially mitigating the negative market reaction associated with the approval of such patents. Tangible assets, by their nature, often involve clear operational procedures and measurable outcomes. OITP principles indicate that organizations can leverage tangible assets to enhance decision quality. In the context of AI-powered service technology patents, tangible assets may serve as reliable benchmarks for evaluating the potential impact on operations and decision-making. This clarity and measurability could positively influence decision quality, aligning with OITP perspectives and potentially leading to a less negative stock market reaction.

Tangibility may influence how investors perceive risks associated with new technologies (Laroche et al., 2003; Marinotti, 2020; Schaefers et al., 2022). The visibility and tangibility of assets can impact risk perceptions within organizations. Firms with higher tangibility may project a sense of stability and concrete operational foundations, potentially mitigating perceived risks associated with the integration of AI in service technology patents. Investors may interpret higher tangibility as a sign of reduced uncertainty, leading to a less negative stock market reaction. Tangible assets, being fundamental to operational processes, can be strategically leveraged in the context of AI-rich service technology patents. Firms with higher tangibility may showcase a strategic approach to integrating AI technologies, utilizing tangible assets as key components in this process. This strategic utilization may contribute to a more positive perception among investors, potentially resulting in a less negative stock market reaction.

In conclusion, from the OITP perspective, the approval of service technology patents featuring high AI scores is posited to have mitigating effects on the negative stock market reactions associated with higher tangibility. Tangible assets play a crucial role in providing visibility, serving as a foundation for adaptation, influencing decision quality, shaping risk perceptions, and facilitating strategic utilization.

Hypothesis 3. For firms with higher tangibility, approval of a service technology patent.

scoring high on AI will be associated with a less negative stock market reaction.

3. Sample and method

3.1. Sample

The present study utilizes four datasets: The Artificial Intelligence Patent Dataset (AIPD), COMPUSTAT data, PATSTAT data, and the KPSS dataset on stock market reaction. The AIPD is a dataset of patents granted between 1976 and 2020, which have been classified into eight AI component technologies using state-of-the-art machine learning-based models (Giczy et al., 2022).[1] The AIPD was merged with COMPUSTAT, OECD PATSTAT data on patent characteristics, and stock market reaction data from the KPSS dataset (Kogan et al., 2017). Manufacturing firms in the present study are defined based on their reported industries within SIC codes ranging from 2000 to 3999. Based on casewise deletion, the final sample includes 759,449 patents, 1306 service technology patents, and 758,143 non-service technology patents from 3899 manufacturing firms from 1980q1 to 2020q4.

3.2. Measures

3.2.1. Stock market reaction

We use patent value estimates derived from Kogan et al.' (2017) work on stock market reactions to patent grants. The stock market reaction signifies the abnormal change in the firm's market capitalization on the day of the patent grant. The dollar value of the stock market reaction is adjusted to 1982 (million) dollars using consumer price index (CPI) values, as per Kogan et al. (2017).

3.2.2. Service technology patents

We use the widely used classification of service technology patents (Geum et al., 2017). The IPC-4 classes from Geum et al. (2017) are listed in Table 1. As service technology patents represent a small portion of our sample, we also conduct a matched-pair sample test.

[1] The AIPD classification models were validated through USPTO patent examiner expert review. Giczy et al. (2022) report model performance metrics for each of the eight AI component technologies, with the methodology achieving strong classification accuracy. Manual expert validation confirms that the machine learning approach reliably identifies AI-related patent content across the technology components most relevant to service applications: machine learning, natural language processing, and knowledge processing. This validation provides confidence that our composite AI score accurately captures the AI-intensity of service technology patents.

3.2.3. AI score of patents

The Artificial Intelligence Patent Dataset (AIPD) employs machine learning techniques to identify patents related to AI technologies (Giczy et al., 2022). For each AI component, a classification model determines if a patent's text and citations represent that technology. This automated patent landscaping approach, based on the work of Abood and Feltenberger (2018), trains models on positive and negative examples to generate component-specific models that then classify the broader patent universe. Manual expert validation from USPTO examiners assesses the performance of these models. The resulting predictions cover over 13 million patents and patent applications from 1976 to 2020.

The AI score of patents is derived from the Artificial Intelligence Patent Dataset (AIPD) developed by Giczy et al. (2022). This dataset employs machine learning techniques to identify and classify patents related to AI technologies across eight components: knowledge processing, speech, hardware, evolutionary computation, natural language processing, machine learning, computer vision, and planning/control. Each AI component captures specific aspects of AI technology. For example, machine learning patents encompass computational models like classification algorithms using labeled training data or neural networks. AI hardware involves physical technologies optimizing AI software, such as Google's Tensor Processing Units bolstering neural networks. Planning/control patents relate to methods for identifying and executing plans to achieve goals under uncertainty, connecting to control theory, simulations, and data systems for tasks like inventory or time management.

In our study, we take the standardized factor score of the AI scores on each of these AI dimensions to create a composite AI score for each patent. A higher AI score indicates a stronger presence of AI-related content and technologies within the patent. This composite score allows us to quantify the degree of AI content across diverse patents, enabling an analysis of AI's role in service technology patents and their market impacts.

3.2.4. Idiosyncratic volatility

Idiosyncratic volatility refers to the unsystematic risk that is independent of overall market risk. It captures the volatility stemming from factors unique to a particular firm rather than due to general market conditions. For example, an airline company may have high idiosyncratic risk related to labor union disputes, aircraft mechanical issues, or fuel price fluctuations that are uncorrelated with broader economic trends.

The Fama-French Carhart (FFC) four-factor model expands on the capital asset pricing model (CAPM) by incorporating additional risk factors beyond just overall market risk (Carhart, 1997). The model specification is shown in eq. (1) below:

$$R_i - R_f = \alpha_i + \beta_{1i}\left(R_m - R_f\right) + \beta_{2i}SMB + \beta_{3i}HML + \beta_{4i}MOM + \varepsilon_i \tag{1}$$

Where: R_i = Return on asset i; Rf = Risk-free rate; $\left(R_m - R_f\right) =) =$ Market risk premium; *SMB* (Small Minus Big) = Size risk factor; *HML* (High Minus Low) = Value risk factor; *MOM* (Momentum) = Momentum risk factor; α_i = Regression intercept; β_{1i} = Asset sensitivity to market risk; β_{2i}, β_{3i}, β_{4i} = Asset sensitivity to SMB, HML, and MOM factors; and ε_i = Residual return.

The FFC model regresses an asset's excess returns against the excess market returns, plus returns on zero-investment factor-mimicking portfolios for size, value, and momentum dimensions. The coefficients quantify asset exposure to each factor as a measure of risk, while α_i captures idiosyncratic return. The element of the Fama-French Carhart four-factor model that captures idiosyncratic volatility is the residual term ε_i. Specifically, ε_i represents the portion of an asset's returns that are not explained by the market, size, value, and momentum risk factors in the model. It quantifies the residual return tied specifically to the individual asset based on its unique attributes.

3.2.5. Tangibility

Tangibility refers to the physical and measurable assets owned by a firm, serving as an indicator of asset realizability (Almeida and Campello, 2007; Villalonga, 2004). Tangibility is the ratio of fixed assets ratio to the Book value of total assets, where fixed assets are the net property, plants, and equipment.

3.2.6. Controls

We include several control variables. Supply chain positions of eigenvector centrality, betweenness centrality, and interconnectedness, although conceptually related, provide distinct insights into node centrality within a network (Borgatti and Li, 2009). Supply chain positions can play a key role in driving servitization efforts. Eigenvector centrality assesses a node's influence or significance based on its connections to highly connected nodes. A node's centrality score relies not only on its direct links but also on the centrality scores of its neighbors. In essence, a node gains importance by associating with other important nodes, reflecting the concept of “prestige” or “influence by association.” High eigenvector centrality indicates nodes connected to other nodes of considerable centrality, signifying their prominent role. Betweenness centrality quantifies a node's bridging role between others in the network by counting the number of shortest paths passing through it, an indicator of a node's potential to facilitate information or resource flow between different parts of the network. Nodes with elevated betweenness centrality act as vital connectors or gatekeepers, enhancing communication and interaction across the network's segments. Interconnectedness measures the number of direct connections a node possesses within the network, focusing solely on the quantity of links rather than their quality or influence. Nodes with high interconnectedness have numerous direct connections to other nodes, providing fundamental insight into a node's reach or popularity within the network. All network variables were standardized.

Based on Borgatti and Li (2009) eigenvector centrality is:

$$E_{it} = \frac{\sum_j I_{ijt} E_{jt}}{\lambda}, \tag{2}$$

Where in eq. (2), $I_{ijt} = 1$ if firms i and j are connected at period t and zero otherwise, for some constant λ (the maximal eigenvalue as in (Bonacich, 1987).

For *betweenness centrality* the measure proposed by Freeman et al. (1979) is as follows:

$$B_{it} = \sum_{j \neq i, i \neq k} \frac{g_{ijkt}}{g_{jkt}}, \tag{3}$$

where in eq. (3), g_{jkt} is the total number of shortest geodesic paths linking firms j and k in year t and g_{ijkt} is the count of the geodesics that contain i in year t. The normalized version is

$$B^*_{it} = \frac{B_{it}}{(N_t - 1)(N_t - 2)/2} \tag{4}$$

where in eq. (4), N_t refers to the network size in year t. We modify Betweenness Centrality as follows:

$$B_{it} = \sum_{j \neq i, i \neq k} \frac{\widetilde{g}_{ijkt}}{\widetilde{g}_{jkt}}, \tag{5}$$

where in eq. (5) $\widetilde{g}_{jkt}$ is the total *length* of shortest geodesic paths linking firms j and k in year t and $\widetilde{g}_{ijkt}$ is the *length* of the shortest geodesic paths that contain i in year t. The “length” is measured using the average culture of firms i, j, and k in year t. We replace it by

$$IC_{it} = \frac{\sum_j \sum_q \widetilde{p}_{iqt}}{n_{it}}, \tag{6}$$

Where in eq. (6), $\widetilde{p}_{iqt}$ is the culture from firm q to i.

Interconnectedness is based on adapting Burt's (1992):

$$IC_{it} = \frac{\sum_j \sum_q p_{iqt} I_{jqt}}{n_{it}}, \quad (7)$$

Where in eq. (7), n_{it} denotes the ego network size of firm i in year t, and $p_{iqt} = n_{it}^{-1}$ (the proportion of node i's costs incurred in maintaining the relationship with node q in period t).

Log of sales is a logged transformation of the firm's total annual sales revenue accounts for company size effects on investor reactions. Larger firms tend to attract greater analyst coverage and media visibility, confounding market responses. *Free cash flow to operating cash flow* is the ratio between free cash flows and cash from operations controls for investment capacity driving investor optimism. Superior liquidity and internal funding enable the pursuit of growth opportunities that manifest in stock gains. Standardizing by operating cash flow benchmarks free cash against current capital requirements. *Debt-to-assets ratio* is a firm's total debt holdings divided by total assets adjusted for capital structure influencing investor perceptions of riskiness and growth prospects. Highly leveraged firms face uncertainty regarding financial constraints in service-related AI inventions.

R&D intensity. Research and development expenditure scaled by sales represents innovative investments that could elicit positive investor reactions. Endogeneity between service AI patents and broader innovation pipelines necessitates controlling R&D intensity known to excite stock markets. *Advertising intensity* is the ratio of advertising expenditures divided by sales, and controls for marketing-based contributors to investor optimism complementary to service AI patents. *Staff-to-sales ratio* is a proxy for the human resource intensity measured by staff expenses per dollar of sales revenue accounts for operational efficiencies simultaneously impacting investor perceptions with service AI automation. Parsing market responses between enhanced productivity and technological capabilities avoids conflating complementary effects.

Finally, we control for firm fixed-effects, state of headquarters fixed-effects, and industry (SIC2) fixed effects.

3.3. Results

We use an OLS specification that controls for firm, industry, and state of headquarters fixed effects. We employ the *-reghdfe-* routine from Stata 17 to estimate the model and use standard errors clustered by industry (SIC2). Table 3 summarizes the descriptive statistics and Pearson correlation coefficients for the variables in our analysis.

In Table 4 we present the estimates. To avoid model misspecification bias based on the selection of controls, we present estimates without controls (models 1–5) and with controls (models 6–11). Hypothesis 1 proposed that approval of a service technology patent scoring high on AI will have a negative stock market reaction. In model 8 in Table 4 and Fig. 1(a), service technology patents with higher AI scores realize a negative market reaction ($\beta = -1.371$, $p < 0.05$).

Hypothesis 2 proposed that for firms with higher idiosyncratic volatility, approval of service technology patents scoring high on AI will be associated with a less negative stock market reaction. In model 9 in Table 4 and Fig. 1(b), service technology patents with higher AI scores realize a less negative market reaction ($\beta = 0.437$, $p < 0.05$).

Hypothesis 3 proposed that for firms with higher tangibility, approval of service technology patents scoring high on AI will be associated with a less negative stock market reaction. In model 10 in Table 4 and Fig. 1(b), we do not find support for the hypothesis ($\beta = -1.358$, $p > 0.10$).

3.4. Robustness checks

3.4.1. Matched pair sample

With service patents representing a much smaller portion of the sample, our significance levels could be influenced by the number of non-service technology patents. Therefore, we use a matched pair sample analysis (Stata 17: *-psmatch2-*, caliper(0.01)). We identify the five nearest neighbors using the predictors and controls as matching variables, along with stock market reaction (to account for patents of comparable value). In Table 5 and Fig. 2, our inferences are consistent with the main inferences, however, with this matched pair sample we find support for all three hypotheses.

3.4.2. Alternate moderators

We use cash conversion cycle, investment intensities (R&D intensity and advertising intensity), efficiency (staff-to-sales and asset turnover), and supply chain network positions (betweenness centrality, eigenvector centrality, and interconnectedness). The cash conversion cycle, a crucial financial metric, can influence how service technology patents with high AI scores influence stock market reactions. If the cash conversion cycle is stretched out, it could mean delayed profits from innovation investments, possibly toning down the initial positive market reaction when patent approvals are announced.

Investment intensities, which cover R&D and advertising spending, add another layer of complexity to stock market reactions. High R&D spending shows a dedication to innovation, but the costs involved may offset immediate positive market feelings. Similarly, while more advertising can boost a company's visibility, it also comes with upfront costs that might affect market reaction.

The staff-to-sales ratio, a measure of employee efficiency, plays a part by showing how a company allocates resources and runs its operations. An efficient staff-to-sales ratio could signal streamlined operations, giving a positive boost to the market when AI-based service patents are revealed. Similarly, higher asset turnover could also signal efficiency and improve market reaction. Supply chain positions of betweenness centrality, eigenvector centrality, and interconnectedness bring in network dynamics that could affect stock market reactions. Companies holding key spots in supply chains might experience stronger market responses due to their connected relationships. On the other hand, less central positions could result in less excitement. The tangled interactions of these supply chain measures make it challenging to make sense of stock market reactions to service technology patents with high AI scores. However, in Table 6, we do not find support for any of these theoretically supported moderators.

3.4.3. Non-linear effects

We further test whether non-linear effects of idiosyncratic volatility and tangibility. Considering diminishing marginal returns for these two moderators provides insight into whether an increase in idiosyncratic volatility has a diminishing impact on the moderation effect, suggesting that the initial rise in volatility may have a more pronounced effect that gradually diminishes. The squared term accommodates potential non-linearities arising from dynamics in these domains, providing a more realistic representation of the relationship under study. Tangibility may have a nonlinear influence on how investors perceive and react to technological innovations. The squared effects of tangibility as a moderator can reveal whether there are critical points at which the relationship experiences qualitative shifts, providing insights into non-linearities in investor responses. There might be an optimal combination of tangible assets and technological innovation for improving market reaction. In Table 7 we do not find significant effects for the nonlinear effects.

3.4.4. Fixed-effects individual slopes

In Table 8, we control for fixed-effects individual slopes for firms, industry, and state of headquarters (Rüttenauer and Ludwig, 2020). The traditional fixed-effects (FE) estimator is susceptible to bias when there are heterogeneous slopes or growth curves associated with the parameter of interest, such as in instances where treatment selection is influenced by individual outcome trajectories. Fixed effects individual slope (FEIS) account for diverse slopes or trends, rendering them a robust tool

Table 3
Descriptives

Summary statistics

	Mean	SD	Min	p25	Median	p75	Max
Stock market reaction (in real terms, standardized)	0.051	1.085	−0.526	−0.489	−0.329	0.024	5.837
Service technology patent	0.002	0.041	0	0	0	0	1
AI score of the patent	−0.068	0.495	−0.307	−0.296	0	0	9.922
Idiosyncratic volatility	0.655	1.156	−0.427	−0.427	0.319	1.401	10.323
Tangibility	0.134	0.181	0	0	0.071	0.195	1.458
Betweenness centrality	0.051	0.905	−1.173	−0.543	0	0.38	5.525
Eigenvector centrality	0.043	1.007	−0.854	−0.617	−0.211	0	4.63
Interconnectedness	0.011	0.826	−1.222	−0.551	0	0.395	4.198
Log of sales	5.344	4.039	0	0	7.2	8.751	10.116
Free cash flow to Operating cash	0.093	8.820	−2999.526	0	0	0.567	2.305
Debt-to-assets	0.24	0.299	0	0	0	0.522	6.883
R&D intensity	0.134	11.328	0	0	0	0.035	6793.75
Advertising intensity	0.005	0.019	0	0	0	0	3.086
Staff-to-sale	0.012	0.309	0	0	0	0	73.309

Pairwise correlations

	Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
1	Stock market reaction (in real terms, standardized)	1.000												
2	Service technology patent	−0.008*	1.000											
3	AI score of the patent	−0.011*	0.003*	1.000										
4	Idiosyncratic volatility	0.008*	−0.024*	−0.038*	1.000									
5	Tangibility	0.308*	0.004*	−0.001	−0.294*	1.000								
6	Betweenness centrality	−0.016*	0.003*	0.000	0.014*	−0.002*	1.000							
7	Eigenvector centrality	−0.013*	0.000	0.002*	0.047*	−0.042*	0.018*	1.000						
8	Interconnectedness	0.030*	0.000	0.000	−0.021*	0.026*	−0.005*	−0.030*	1.000					
9	Log of sales	0.141*	−0.020*	−0.017*	−0.415*	0.497*	−0.023*	−0.104*	0.015*	1.000				
10	Free cash flow to Operating cash	0.014*	0.000	−0.001	−0.005*	0.005*	−0.001	−0.001	−0.002	0.017*	1.000			
11	Debt-to-assets	0.220*	0.000	−0.002	−0.219*	0.497*	0.008*	−0.073*	0.021*	0.495*	0.014*	1.000		
12	R&D intensity	−0.003*	0.000	0.000	−0.004*	−0.003*	0.000	−0.001	−0.001	−0.010*	−0.001	0.005*	1.000	
13	Advertising intensity	0.286*	0.001	−0.004*	−0.122*	0.168*	−0.024*	−0.013*	0.036*	0.194*	0.010*	0.285*	−0.002	1.000
14	Staff-to-sale	0.008*	0.003*	0.002	−0.024*	0.037*	0.003*	−0.002	0.001	0.010*	−0.001	0.024*	0.143*	0.023*

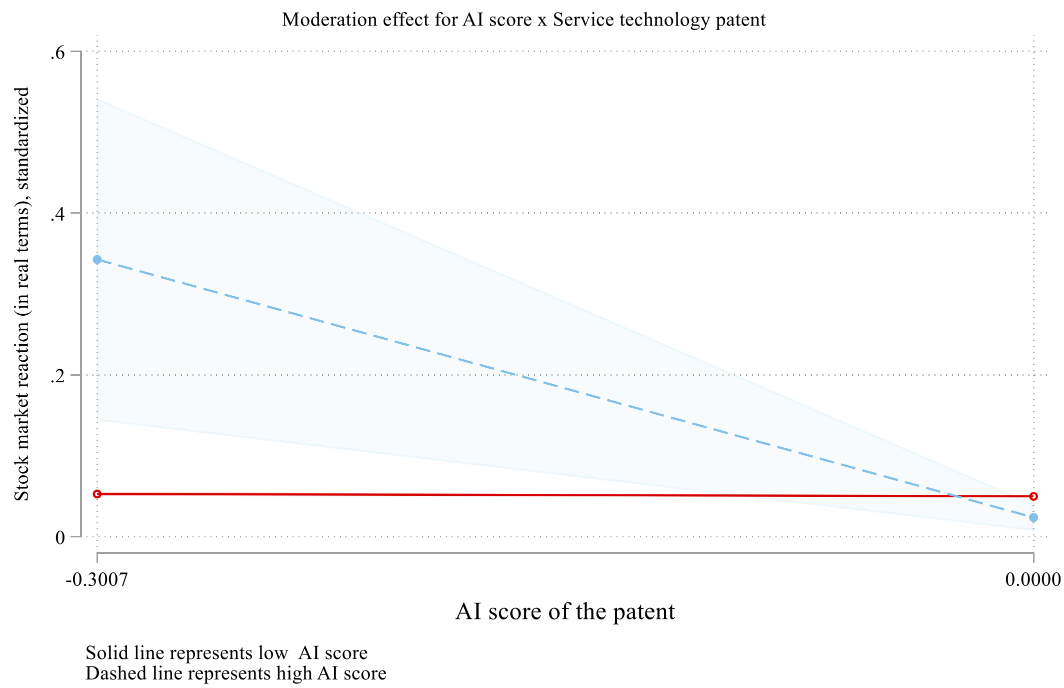
Notes
$N = 759{,}449$; 1306 service technology patents and 758,143 non-service technology patents from 3899 manufacturing firms from 1980q1 to 2020q4; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4
Main analysis estimates.

			Stock market reaction (in real terms), standardized										
			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Service technology patent			−0.0738**	−0.117***	−0.0576**	−0.00948	0.0511		−0.0199	−0.0577**	−0.0221	0.00863	0.0492
			(0.0302)	(0.0323)	(0.0271)	(0.0235)	(0.0322)		(0.0207)	(0.0265)	(0.0293)	(0.0252)	(0.0298)
AI score of the patent				−0.0261***	−0.0383**	−0.00749	−0.0147			−0.0223***	−0.0326**	−0.00605	−0.0110
				(0.00861)	(0.0141)	(0.00704)	(0.0132)			(0.00719)	(0.0128)	(0.00546)	(0.0118)
Service technology patent x AI score of the patent	**[H1: -]**	Supported		−1.581***	−1.466***	−1.213***	−1.248***			−1.371***	−1.414***	−1.170***	−1.466***
				(0.427)	(0.471)	(0.361)	(0.432)			(0.392)	(0.450)	(0.356)	(0.491)
Idiosyncratic volatility					0.119**		0.120**				0.0764**		0.0781**
					(0.0546)		(0.0552)				(0.0355)		(0.0358)
Service technology patent x Idiosyncratic volatility					0.0740**		0.0733*				0.0687**		0.0674**
					(0.0311)		(0.0411)				(0.0265)		(0.0316)
AI score of the patent x Idiosyncratic volatility					0.0182**		0.0108				0.0139*		0.00714
					(0.00858)		(0.00750)				(0.00763)		(0.00678)
Service technology patent x AI score of the patent x Idiosyncratic volatility	**[H2: +]**	Supported			0.417***		0.455***				0.437***		0.512***
					(0.136)		(0.152)				(0.142)		(0.159)
Tangibility						−0.0157	0.0686					0.264	0.296
						(0.501)	(0.462)					(0.365)	(0.352)
Service technology patent x Tangibility						−0.727**	−0.753**					−0.501*	−0.537*
						(0.343)	(0.332)					(0.271)	(0.275)
AI score of the patent x Tangibility						−0.153***	−0.138***					−0.135***	−0.127***
						(0.0287)	(0.0325)					(0.0174)	(0.0230)
Service technology patent x AI score of the patent x Tangibility	**[H3: +]**	Not supported				−2.412	−1.928					−1.358	−0.580
						(1.485)	(1.179)					(1.369)	(1.157)
Betweenness centrality								6.10e-05	6.11e-05	5.30e-05	−0.000508	0.000483	−5.60e-05
								(0.00517)	(0.00517)	(0.00517)	(0.00501)	(0.00565)	(0.00549)
Eigenvector centrality								0.00114	0.00114	0.00115	0.00134	0.00120	0.00141
								(0.00130)	(0.00130)	(0.00130)	(0.00131)	(0.00136)	(0.00137)
Interconnectedness								0.00314	0.00314	0.00316	0.00319	0.00281	0.00280
								(0.00379)	(0.00379)	(0.00379)	(0.00390)	(0.00407)	(0.00416)
Log of sales								0.233*	0.233*	0.232*	0.213*	0.238*	0.218*
								(0.122)	(0.122)	(0.122)	(0.115)	(0.124)	(0.117)
Free cash flow to Operating cash flow to current liabilities								0.000356**	0.000356**	0.000355**	0.000353*	0.000370**	0.000370**
								(0.000160)	(0.000160)	(0.000160)	(0.000176)	(0.000160)	(0.000173)
Debt-to-assets								−0.114	−0.114	−0.113	−0.101	−0.0950	−0.0797
								(0.156)	(0.156)	(0.155)	(0.146)	(0.154)	(0.146)
R&D intensity								0.000163*	0.000163*	0.000163*	0.000146**	0.000169*	0.000153*
								(7.92e-05)	(7.91e-05)	(7.87e-05)	(6.97e-05)	(8.58e-05)	(7.57e-05)
Advertising intensity								3.927	3.927	3.928	4.125	3.919	4.121
								(4.276)	(4.276)	(4.274)	(4.317)	(4.305)	(4.348)
Staff-to-sale								−0.0276	−0.0276	−0.0274	−0.0238	−0.0275	−0.0239
								(0.0491)	(0.0490)	(0.0488)	(0.0465)	(0.0489)	(0.0464)
Firm fixed-effects			Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
State fixed-effects			Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
SIC (2-digit) fixed-effects			Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
s.e. clustered by			sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2
Constant			0.0506***	0.0489***	−0.0290	0.0508	−0.0386	−1.185*	−1.185*	−1.185*	−1.133*	−1.253*	−1.209*
			(5.19e-05)	(0.000566)	(0.0360)	(0.0669)	(0.0767)	(0.667)	(0.667)	(0.667)	(0.655)	(0.691)	(0.678)
Observations			759,449	759,449	759,449	759,449	759,449	759,449	759,449	759,449	759,449	759,449	759,449
R-squared			0.517	0.518	0.522	0.518	0.523	0.533	0.533	0.533	0.535	0.534	0.536

Notes
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

(a). Moderation plot [H1]



(b). Moderation plot, main analysis [H2 and H3]

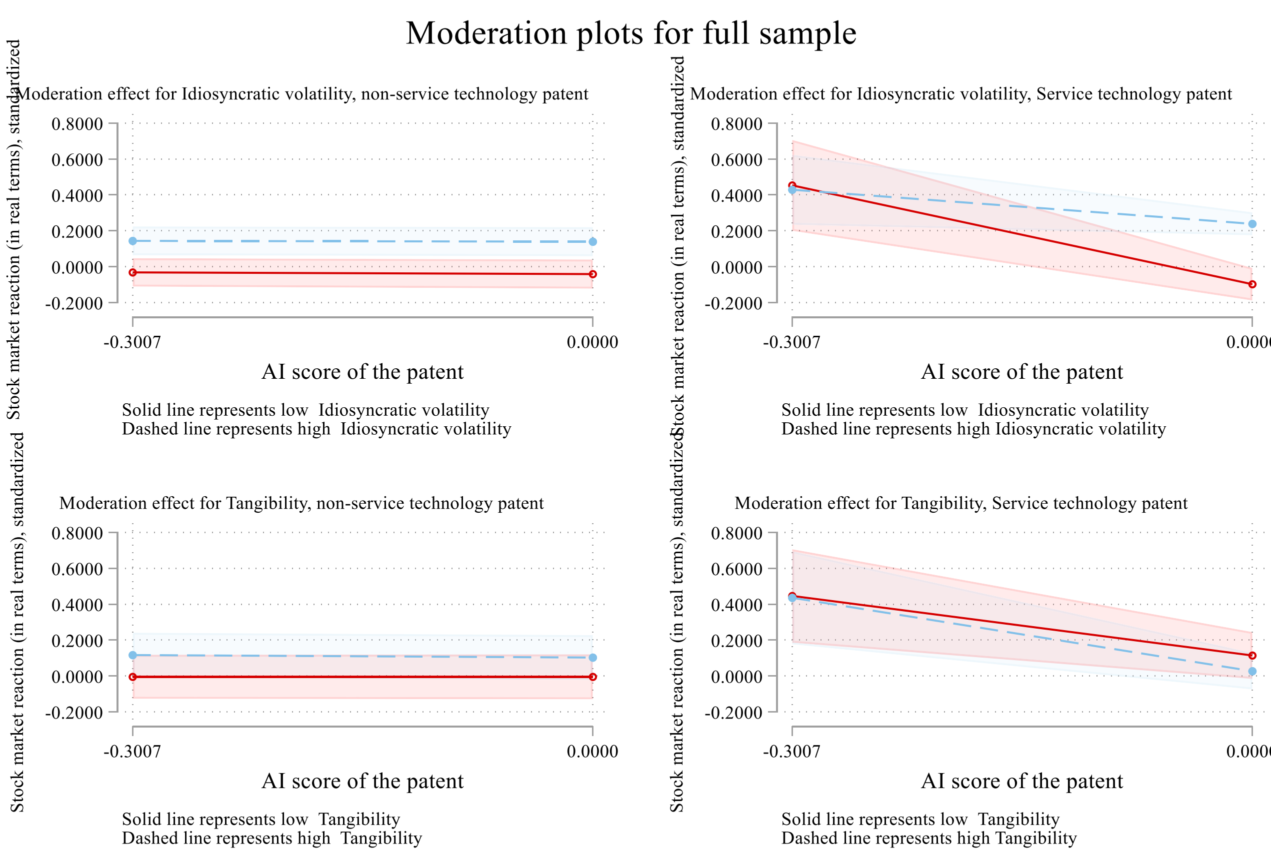


Fig. 1. Moderation plot for the main analysis.
Fig. 1(a). Moderation plot [H1].
Fig. 1(b). Moderation plot, main analysis [H2 and H3].

Table 5
Matched pair sampling.

	DV = Stock market reaction (in real terms), standardized	
	(1)	(2)
Service technology patent	0.0363**	0.0825***
	(0.00838)	(0.00908)
AI score of the patent	−0.00309	−0.0779***
	(0.00147)	(0.00150)
Service technology patent x AI score of the patent	−1.317***	−1.057***
	(0.0410)	(0.0802)
Idiosyncratic volatility		0.0659***
		(0.0102)
Tangibility		0.694***
		(0.0655)
Service technology patent x Idiosyncratic volatility		0.0759***
		(0.00511)
Service technology patent x Tangibility		−0.241***
		(0.0216)
AI score of the patent x Idiosyncratic volatility		0.0254***
		(0.00262)
AI score of the patent x Tangibility		0.176***
		(0.00274)
Service technology patent x AI score of the patent x Idiosyncratic volatility		0.238**
		(−0.0892)
Service technology patent x AI score of the patent x Tangibility		−0.672***
		(0.143)
Controls	Included	Included
Firm fixed-effects	Included	Included
State fixed-effects	Included	Included
SIC (2-digit) fixed-effects	Included	Included
s.e. clustered by	sic2	sic2
Constant	−0.627***	−0.696***
	(0.00504)	(0.0277)
Observations	7225	7225
R-squared	0.519*	0.531

Notes.
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

for the analysis of panel data and other forms of multilevel data. In Table 8 and Fig. 3, our inferences are consistent with those of the main inferences.

3.4.5. Analysis by decade

To address potential temporal heterogeneity in the relationship between AI-based service technology patents and market reactions, we conducted a decade-wise analysis from the 1980s through the 2020s (Table 9). This analysis reveals an evolution of the effects observed in our main specification. The interaction between service technology patents and AI score (H1) shows an interesting pattern across decades. While consistently negative, the effect becomes statistically significant only from 2010 onward, with the magnitude increasing over time. This suggests that the market's negative reaction to AI-intensive service patents has intensified as AI technologies have become more prevalent and perhaps more disruptive to existing business models. The moderating effect of idiosyncratic volatility (H2) also exhibits a temporal pattern. The positive moderation effect is not significant in the 1980s, becomes marginally significant in the 1990s, and strengthens in subsequent decades. This evolution may reflect the market's growing understanding of how firm-specific risk factors interact with AI-based innovations in services. Interestingly, the moderating effect of tangibility (H3) remains non-significant across all decades, consistent with our main findings. However, the direct effect of tangibility on AI score becomes increasingly negative and significant over time, suggesting that the market increasingly values intangible assets in the context of AI-based service innovations. These decade-wise results suggest that the effects we observe in our main specification have strengthened over time with effects robustly becoming significant after the 2010s, possibly reflecting the increasing importance and understanding of AI in service innovations.

The decade-wise analysis reveals that negative market reactions to AI service patents emerged and strengthened over time. In the 1980s–1990s, when AI patents were rare and investor evaluation frameworks underdeveloped, market reactions were noisy and non-significant. The 2000s show marginal significance emerging, while the 2010s–2020s demonstrate robust negative effects. This temporal strengthening aligns with institutional learning about AI implementation realities. As AI moved from theoretical promise to practical deployment in the 2010s, markets gained experience with the gap between AI capabilities and realized servitization value. The dot-com crash, early AI winter disappointments, and visible failures of AI implementation projects likely shaped this learned skepticism. Paradoxically, as AI technologies matured and became more prevalent, investor caution intensified rather than diminished—suggesting markets have become more sophisticated in recognizing implementation challenges specific to AI servitization in manufacturing contexts. The idiosyncratic volatility moderation effect also emerging in recent decades suggests that only after markets developed frameworks for evaluating AI risks did firm-specific uncertainty characteristics become relevant moderating factors.

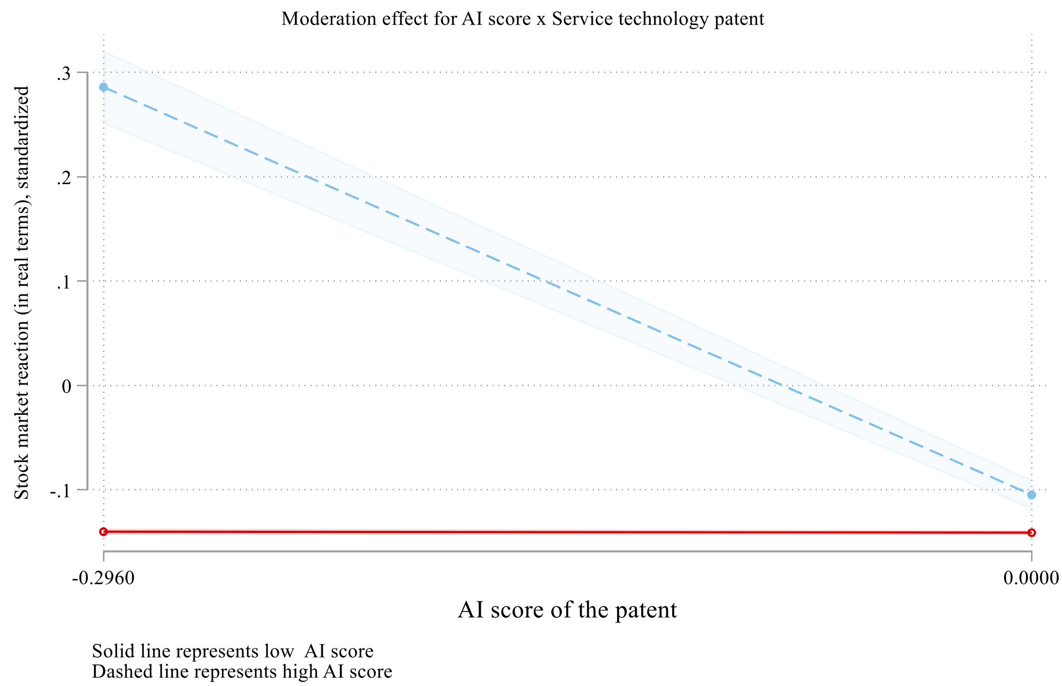
4. Discussion

Servitization represents a strategic shift for manufacturers toward integrated offerings bundling physical goods and services (Faramarzi et al., 2023; Vandermerwe and Rada, 1988). However, despite potential advantages, embedding AI within service patents poses implementation hurdles requiring mitigation so as not to deter servitization initiatives. Introducing AI service patents layers additional complexity upon manufacturers with finite cognitive bandwidth, potentially overwhelming their capacity to process information changes (Galbraith, 1977; Joseph and Gaba, 2020). Complications may also arise from ambiguities or conflicts in enforcing AI patent protections still nascent in application. Ultimately, these challenges manifest in negative market reactions to AI service patent approvals, reflecting perceived threats to successfully servitizing. Yet certain factors affecting a firm's ability to absorb and govern complexity may reduce downside sensitivity, as explored through an empirical lens in this research. The inquiry aims to deliver a balanced and generalizable perspective to inform manufacturers pursuing AI-enabled service innovation.

Leveraging insights from transaction cost economics and the organizational information processing perspective, the findings of our study challenge the anticipated benefits of servitization by asserting that the stock market reactions to the approval of service technology patents with high AI scores are negatively perceived (Brax et al., 2021; Dmitrijeva et al., 2022; Kohtamäki et al., 2020). However, we find that idiosyncratic volatility acts as a mitigating factor, alleviating the adverse reactions, while tangibility does not demonstrate a similar effect. Robustness analyses, including matched-pair sampling, assessment of non-linear effects, and consideration of time-trends for firm, industry, and state of headquarters, consistently support these findings. Despite exploring alternate moderators, such as cash conversion cycle, investment intensities, efficiency, and supply chain positions, including betweenness centrality, eigenvector centrality, and interconnectedness, our results underscore the specificity of the identified hypotheses.

The strengthening negative effect of AI-intensive service patents on market reactions from the 2010s to the 2020s (H1), but not in the earlier decades aligns with the evolving understanding of AI's role in servitization. This trend can be interpreted through the lens of the Technology Acceptance Model (TAM) (Davis (Vaittinen and Nenonen, 2018), 1989) and its extensions. In earlier decades, when AI was in its infancy, the market's perception of its usefulness and ease of use in service contexts

(a). Moderation plots (H1)



(b). Moderation plots (H2 and H3)

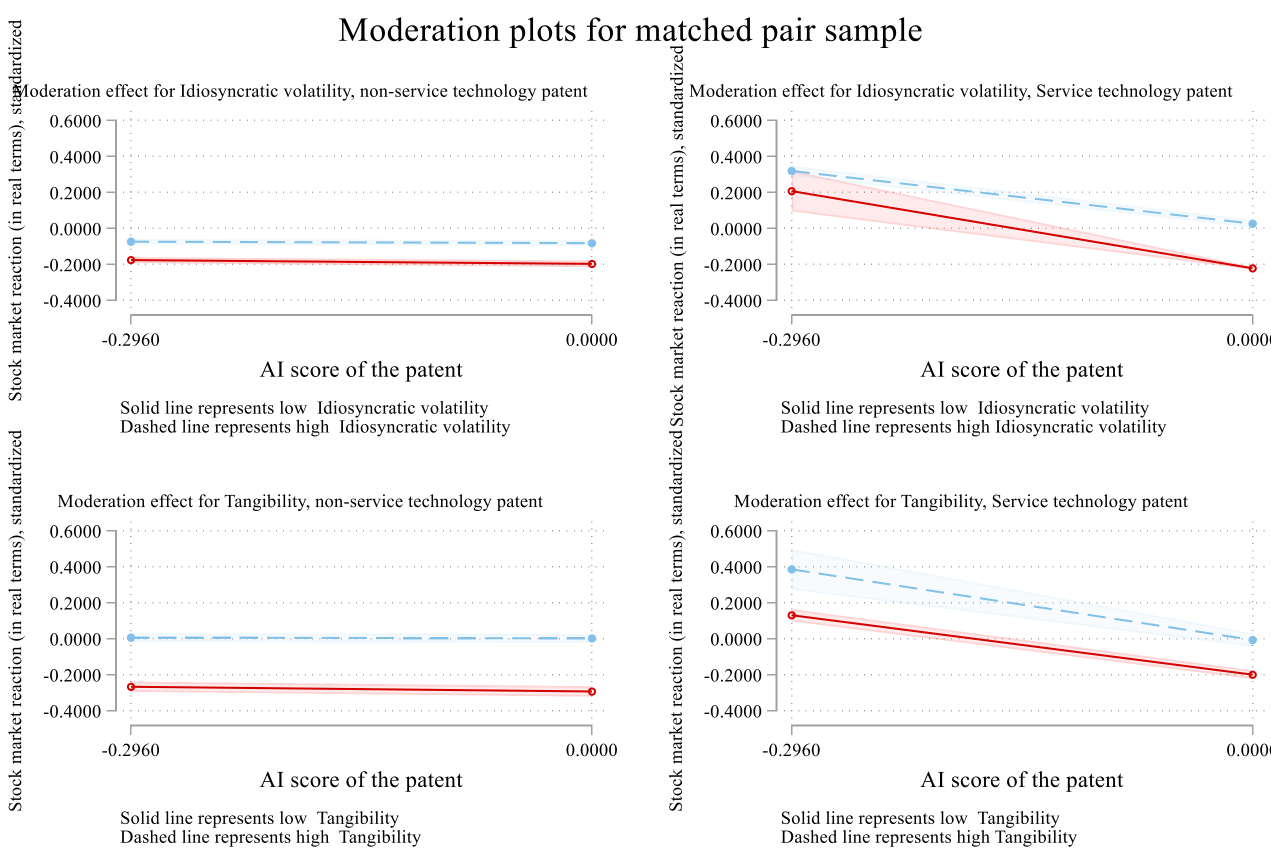


Fig. 2. Moderation plot, matched pair sample.
Fig. 2(a). Moderation plots (H1).
Fig. 2(b). Moderation plots (H2 and H3).

Table 6
Alternate moderators.

	DV = Stock market reaction (in real terms), standardized							
Moderator	Cash conversion cycle	R&D intensity	Advertising intensity	Staff to sale	Asset turnover	Betweenness centrality	Eigenvector centrality	Interconnectedness
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Service technology patent	−0.0776	−0.0544*	−0.0229	−0.0230	−0.00954	−0.0608**	−0.0576**	−0.0577**
	(0.0485)	(0.0283)	(0.0363)	(0.0163)	(0.0189)	(0.0271)	(0.0264)	(0.0264)
AI score of the patent	−0.0295*	−0.0224***	−0.0217***	−0.0221***	0.00330	−0.0223***	−0.0224***	−0.0223***
	(0.0161)	(0.00720)	(0.00721)	(0.00715)	(0.00911)	(0.00721)	(0.00718)	(0.00717)
Service technology patent x AI score of the patent	−0.913	−1.734***	−1.283***	−1.196***	−0.997**	−1.326***	−1.358***	−1.370***
	(0.816)	(0.461)	(0.397)	(0.337)	(0.391)	(0.292)	(0.385)	(0.394)
Log of sales	0.310**	0.232*	0.232*	0.232*	0.215*	0.232*	0.232*	0.232*
	(0.144)	(0.122)	(0.122)	(0.122)	(0.116)	(0.122)	(0.122)	(0.122)
Free cash flow to Operating cash flow to current liabilities	0.000283*	0.000355**	0.000355**	0.000354**	0.000351**	0.000355**	0.000355**	0.000355**
	(0.000152)	(0.000159)	(0.000159)	(0.000160)	(0.000161)	(0.000160)	(0.000160)	(0.000160)
Debt-to-assets	−0.196	−0.114	−0.113	−0.114	−0.0936	−0.113	−0.113	−0.113
	(0.161)	(0.155)	(0.155)	(0.155)	(0.144)	(0.155)	(0.155)	(0.155)
R&D intensity	0.000155**	0.000175**	0.000162*	0.000158**	0.000112	0.000163*	0.000163*	0.000163*
	(6.54e-05)	(8.03e-05)	(7.87e-05)	(7.38e-05)	(7.28e-05)	(7.86e-05)	(7.87e-05)	(7.87e-05)
Advertising intensity	4.219	3.928	3.913	3.935	3.964	3.928	3.928	3.928
	(4.531)	(4.274)	(4.219)	(4.275)	(4.262)	(4.274)	(4.274)	(4.274)
Staff-to-sale	−0.0161	−0.0274	−0.0274	−0.0459	−0.0164	−0.0274	−0.0274	−0.0274
	(0.0434)	(0.0488)	(0.0488)	(0.0632)	(0.0423)	(0.0488)	(0.0488)	(0.0489)
Betweenness centrality	−0.000122	5.26e-05	5.71e-05	4.65e-05	−6.26e-05	4.65e-05	5.33e-05	5.52e-05
	(0.00526)	(0.00517)	(0.00517)	(0.00517)	(0.00531)	(0.00504)	(0.00517)	(0.00517)
Eigenvector centrality	0.00123	0.00115	0.00114	0.00115	0.00108	0.00115	0.00126	0.00115
	(0.00134)	(0.00130)	(0.00130)	(0.00130)	(0.00126)	(0.00130)	(0.00127)	(0.00130)
Interconnectedness	0.00358	0.00316	0.00316	0.00317	0.00332	0.00316	0.00316	0.00298
	(0.00373)	(0.00379)	(0.00379)	(0.00379)	(0.00378)	(0.00379)	(0.00379)	(0.00378)
Moderator	4.43e-07				−0.181			
	(1.95e-05)				(0.117)			
Service technology patent x Moderator	0.000317	−0.328	−6.039**	−1.004**	−0.0556	0.0266*	−0.00244	0.000669
	(0.000288)	(0.565)	(2.469)	(0.367)	(0.0638)	(0.0137)	(0.00856)	(0.0147)
AI score of the patent x Moderator	5.47e-07	6.88e-05**	−0.131	−0.136	−0.0750*	0.000729	0.00144	−0.00283
	(9.30e-05)	(2.64e-05)	(0.440)	(0.111)	(0.0417)	(0.00378)	(0.00153)	(0.00212)
Service technology patent x AI score of the patent x Moderator	−0.00497	10.08*	−14.17	−7.809***	−0.474	−0.148	0.270***	0.0486
	(0.00679)	(5.278)	(12.19)	(2.206)	(0.721)	(0.550)	(0.0832)	(0.492)
Controls	Included	Included	Included	Included	Included	Included	Included	Included
Firm fixed-effects	Included	Included	Included	Included	Included	Included	Included	Included
State fixed-effects	Included	Included	Included	Included	Included	Included	Included	Included
SIC (2-digit) fixed-effects	Included	Included	Included	Included	Included	Included	Included	Included
s.e. clustered by	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2
Constant	−1.172*	−1.185*	−1.184*	−1.182*	−1.028	−1.185*	−1.185*	−1.185*
	(0.639)	(0.667)	(0.667)	(0.668)	(0.636)	(0.667)	(0.667)	(0.667)
Observations	602,367	759,449	759,449	759,449	759,449	759,449	759,449	759,449
R-squared	0.520	0.533	0.533	0.533	0.534	0.533	0.533	0.533

Notes
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

was likely limited, resulting in muted reactions. The increasing moderating effect of idiosyncratic volatility (H2) over the 2010s and 2020s indicates that as AI technologies matured, firms with higher idiosyncratic volatility - often associated with greater flexibility and adaptability - appear better positioned to leverage AI in their service innovations, suggesting an evolving complementarity between firm-specific risk factors and technological capabilities, reflecting the growing importance of dynamic capabilities in rapidly changing technological landscapes (Teece et al., 1997). The persistent non-significance of tangibility as a moderator (H3) across decades, coupled with its increasingly negative interaction with AI score, provides insights into the changing nature of valuable resources in the digital age. Aligning with the knowledge-based view of the firm (Grant, 2006), this finding suggests that as AI technologies advance, intangible assets such as data, algorithms, and AI expertise become increasingly critical for successful servitization, potentially diminishing the relative importance of tangible assets. Our decade-wise analysis reveals that the relationships between AI, service innovation, and market reactions are not static but evolve with technological advancements and collective learning, and findings underscore the importance of considering temporal dynamics in understanding the complex interplay between technological innovation, firm characteristics, and market perceptions in the context of servitization.

Table 7
Squared term estimates.

	DV = Stock market reaction (in real terms), standardized		
	(1)	(2)	(3)
Service technology patent	−0.0261***	0.00367	0.0513
	(0.00861)	(0.0241)	(0.0374)
AI score of the patent	−0.00976	0.00514	0.0161***
	(0.00575)	(0.00626)	(0.00512)
Service technology patent x AI score of the patent	−1.051***	−0.798	−1.724*
	(0.342)	(0.514)	(0.895)
Idiosyncratic volatility		0.0497*	0.0792
		(0.0257)	(0.0904)
Service technology patent x Idiosyncratic volatility		0.0579**	0.0983
		(0.0217)	(0.0676)
AI score of the patent x Idiosyncratic volatility		0.00222	0.0301***
		(0.00344)	(0.00875)
Service technology patent x AI score of the patent x Idiosyncratic volatility		0.308**	0.193
		(0.124)	(0.615)
Idiosyncratic volatility-squared			−0.0159
			(0.0300)
Service technology patent x Idiosyncratic volatility-squared			−0.0218
			(0.0354)
AI score of the patent x Idiosyncratic volatility-squared			−0.0139***
			(0.00337)
Service technology patent x AI score of the patent x Idiosyncratic volatility-squared			0.157
			(0.287)
Tangibility		−1.403	1.773***
		(1.296)	(0.208)
Service technology patent x Tangibility		−0.0123	−0.776***
		(0.273)	(0.230)
AI score of the patent x Tangibility		−0.114***	−0.243***
		(0.0306)	(0.0516)
Service technology patent x AI score of the patent x Tangibility		−1.515	5.945
		(0.890)	(9.714)
Tangibility-squared			−3.578***
			(0.138)
Service technology patent x Tangibility-squared			1.356*
			(0.653)
AI score of the patent x Tangibility-squared			0.138***
			(0.0389)
Service technology patent x AI score of the patent x Tangibility-squared			−13.30
			(20.98)
Controls	Included	Included	Included
Firm fixed-effects	Included	Included	Included
State fixed-effects	Included	Included	Included
SIC (2-digit) fixed-effects	Included	Included	Included
s.e. clustered by	sic2	sic2	sic2
Constant	−0.696**	−0.553*	−0.544*
	(0.289)	(0.289)	(0.272)
Observations	759,449	759,449	759,449
R-squared	0.631	0.636	0.651

Notes.
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

Table 8
Fixed-effects individual slopes estimates.

	DV = Stock market reaction (in real terms), standardized	
	(1)	(2)
Service technology patent	−0.0261***	0.00367
	(0.00861)	(0.0241)
AI score of the patent	−0.00976	0.00514
	(0.00575)	(0.00626)
Service technology patent x AI score of the patent	−1.051***	−0.798
	(0.342)	(0.514)
Idiosyncratic volatility		0.0497*
		(0.0257)
Tangibility		−1.403
		(1.296)
Service technology patent x Idiosyncratic volatility		0.0579**
		(0.0217)
Service technology patent x Tangibility		−0.0123
		(0.273)
AI score of the patent x Idiosyncratic volatility		0.00222
		(0.00344)
AI score of the patent x Tangibility		−0.114***
		(0.0306)
Service technology patent x AI score of the patent x Idiosyncratic volatility		0.308**
		(0.124)
Service technology patent x AI score of the patent x Tangibility		−1.515
		(0.890)
Controls	Included	Included
Firm fixed-effects	Included	Included
State fixed-effects	Included	Included
SIC (2-digit) fixed-effects	Included	Included
s.e. clustered by	sic2	sic2
Constant	−0.696**	−0.553*
	(0.289)	(0.289)
Observations	759,449	759,449
R-squared	0.631	0.636

Notes.
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

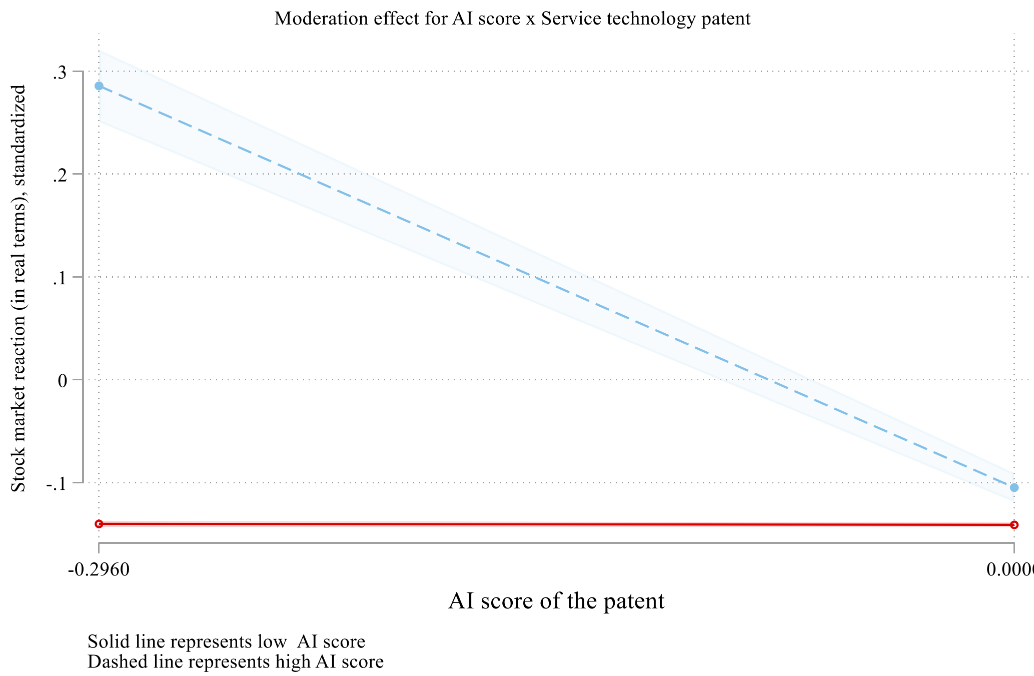
4.1. Theoretical implications

The findings of this study contribute to the servitization literature in several ways by introducing new perspectives on the relationship between servitization, AI-enabled service innovation, and firm performance.

First, in contrast to the widely held view that servitization is inherently beneficial for manufacturing firms, the study's findings suggest that the pursuit of service innovation, particularly in the context of AI-enabled service technology, can lead to paradoxical outcomes (Brax et al., 2021; Dmitrijeva et al., 2022; Kohtamäki et al., 2020). The observed negative stock market reactions upon the approval of high AI-scoring service technology patents indicate that investors, at least in the short-run, may perceive such innovations as potential threats to the existing business models and value propositions of manufacturing firms. This finding aligns with the paradox theory of servitization, which suggests that manufacturing firms may face tensions and contradictions when attempting to balance their traditional product-centric business models with the emerging service-oriented approaches (Abou-Foul et al., 2023; Giczy et al., 2022; Sjödin et al., 2023). AI-enabled service innovations can disrupt existing product-based revenue streams and create uncertainties about the future direction of the firm, leading to investor skepticism and negative market reactions (Akter et al., 2023).

Second, one of the key findings of the study is the identification of idiosyncratic volatility, a measure of firm-specific risk, as a mitigating factor against the negative stock market reactions associated with service technology patents with high AI scores. Firms with higher idiosyncratic volatility, typically characterized by greater agility and responsiveness to market changes, may be better equipped to navigate the potential disruptions caused by service technology innovations (Dai, 2021; Mazzucato and Tancioni, 2008). Firms with higher idiosyncratic volatility are often characterized by a more decentralized decision-

(a). Moderation effects (H1), for fixed-effects individual slopes



(b). Moderation plots for fixed-effects individual slopes

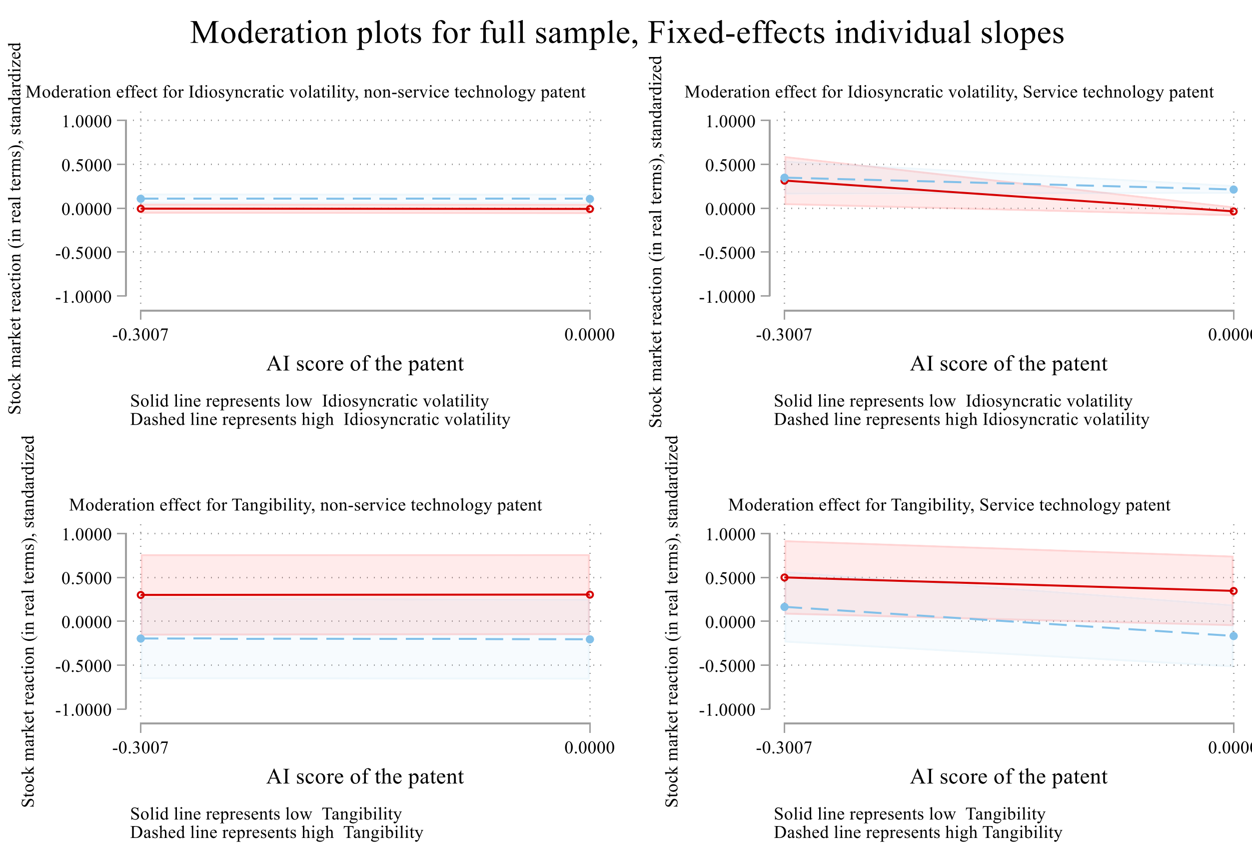


Fig. 3. Moderation plots.
Fig. 3(a). Moderation effects (H1), for fixed-effects individual slopes.
Fig. 3(**b).** Moderation plots for fixed-effects individual slopes.

Table 9
Main specification analyses, by decade.

		Full sample (main analysis)		1980s		1990s		2000s		2010s		2020s	
		(1)	(2)										
Service technology patent		−0.0577**	0.0492	−0.0102	0.0183	−0.0231	0.0275	−0.0412*	0.0381	−0.0349	0.0437	−0.0501*	0.0468
		(0.0265)	(0.0298)	(0.041)	(0.046)	(0.036)	(0.039)	(0.030)	(0.032)	(0.031)	(0.031)	(0.029)	(0.030)
AI score of the patent		−0.0223***	−0.0110	−0.0051	−0.0037	−0.0112	−0.0068	−0.0189**	−0.0095	−0.0155*	−0.0103	−0.0208**	−0.0107
		(0.00719)	(0.0118)	(0.010)	(0.013)	(0.009)	(0.012)	(0.008)	(0.012)	(0.008)	(0.012)	(0.008)	(0.012)
Service technology patent x AI score of the patent	**[H1: -]**	−1.371***	−1.466***	−0.328	−0.415	−0.673	−0.781*	−1.015	−1.198**	−0.889**	−1.321***	−1.252**	−1.437***
		(0.392)	(0.491)	(0.589)	(0.673)	(0.512)	(0.598)	(0.758)	(0.521)	(0.334)	(0.502)	(0.427)	(0.487)
Idiosyncratic volatility			0.0781**		0.0231		0.0412		0.0587*		0.0524		0.0693*
			(0.0358)		(0.041)		(0.039)		(0.037)		(0.038)		(0.037)
Service technology patent x Idiosyncratic volatility			0.0674**		0.0198		0.0351		0.0498*		0.0442		0.0591*
			(0.0316)		(0.039)		(0.036)		(0.033)		(0.034)		(0.033)
AI score of the patent x Idiosyncratic volatility			0.00714		0.00201		0.00389		0.00561		0.00492		0.00651
			(0.00678)		(0.008)		(0.008)		(0.007)		(0.007)		(0.007)
Service technology patent x AI score of the patent x Idiosyncratic volatility	**[H2: +]**		0.512***		0.145		0.261		0.389		0.342**		0.468**
			(0.159)		(0.198)		(0.194)		(0.568)		(0.170)		(0.165)
Tangibility			0.296		0.0871		0.158		0.231		0.203		0.271
			(0.352)		(0.412)		(0.389)		(0.371)		(0.378)		(0.363)
Service technology patent x Tangibility			−0.537*		−0.159		−0.285		−0.421*		−0.368		−0.492*
			(0.275)		(0.332)		(0.308)		(0.289)		(0.296)		(0.286)
AI score of the patent x Tangibility			−0.127***		−0.0372		−0.0678*		−0.0998**		−0.0871*		−0.118**
			(0.0230)		(0.029)		(0.027)		(0.025)		(0.025)		(0.024)
Service technology patent x AI score of the patent x Tangibility	**[H3: +]**		−0.580		−0.171		−0.312		−0.458		−0.401		−0.535
			(1.157)		(1.389)		(1.281)		(1.212)		(1.242)		(1.189)
Controls		Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
Firm fixed-effects		Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
State fixed-effects		Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
SIC (2-digit) fixed-effects		Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included	Included
s.e. clustered by		sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2	sic2
Constant		−1.185*	−1.209*	−0.893	−0.921	−1.021	−1.058	−1.137*	−1.172*	−1.201*	−1.198*	−1.215*	−1.211*
		(0.667)	(0.678)	(0.812)	(0.825)	(0.745)	(0.761)	(0.701)	(0.712)	(0.673)	(0.681)	(0.669)	(0.675)
Observations		759,449	759,449	22,783	22,783	68,350	68,350	151,890	151,890	303,780	303,780	212,646	212,646
R-squared		0.533	0.536	0.412	0.415	0.458	0.461	0.501	0.504	0.528	0.531	0.534	0.537

Notes
*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$.

making structure, empowered employees, and a culture of experimentation, allowing them to respond quickly and effectively to changing market conditions (Ross et al., 2018).

Third, the study's findings indicate that tangibility, a measure of the physical nature of a firm's assets, does not mitigate the negative stock market reactions associated with service technology patents with high AI scores (Atasoy et al., 2022; Jancenelle, 2021; Laroche et al., 2003; Schaefers et al., 2022). This suggests that the potential disruptions caused by AI-enabled service innovations are not limited to firms with high tangible assets, as even firms with intangible assets may face challenges in adapting to the evolving servitization landscape (Almeida and Campello, 2007). The finding contrasts with the prevailing notion that firms with higher tangibility may be more resistant to the disruptive effects of AI-enabled service innovations. While tangibility can provide a competitive advantage in traditional manufacturing industries, it can also hinder a firm's ability to adapt to the rapidly changing digital landscape and embrace new service-oriented business models (Gebauer et al., 2021).

Fifth, related to the research directions outlined in Faramarzi et al. (2023) meta-analysis, we address the following calls for research. The study's findings highlight the importance of considering service typology when assessing the impact of servitization on firm performance. The observation that firms with high AI scores on service technology patents experience negative stock market reactions suggests that investors may perceive certain types of service innovations as potential threats to the existing business models and value propositions of manufacturing firms. The study's findings align with this research direction by emphasizing the potential for smart technologies, particularly AI, to disrupt traditional manufacturing business models and create uncertainties about the future direction of the firm. The negative stock market reactions observed upon the approval of high AI-scoring service technology patents suggest that investors may be skeptical about the ability of manufacturing firms to navigate the complexities of AI-enabled service innovation and adapt to the evolving servitization landscape. This finding highlights the importance of further research to investigate the potential benefits and challenges associated with AI-enabled service innovation for manufacturing firms. Related to additional research questions on generating more insights into mediating and moderating mechanisms. The study's findings contribute to this research direction by identifying idiosyncratic volatility as a potential moderating factor in the relationship between AI-enabled service innovation and firm performance.

4.2. Practical implications

For manufacturing executives, the negative short-term market reactions we document indicate that AI service patent announcements should be treated as strategically significant investor-relations events rather than routine legal milestones. Firms need to manage these events proactively along several dimensions. First, announcements should ideally follow demonstrable progress rather than precede it. Establishing proof-of-concept through pilot programs, early customer deployments, or internal performance evidence can help alleviate investor concerns about implementation feasibility and execution risk. Second, patent announcements should be embedded within a broader capability narrative. Communicating (a) the specific organizational capabilities that undergird the patent (e.g., specialized AI talent, data infrastructure, and strategic partnerships), (b) the concrete ways in which the innovation lowers transaction costs or alleviates salient customer pain points, and (c) a realistic implementation roadmap with intermediate milestones can reduce the uncertainty that fuels negative valuation responses. Third, our finding that higher idiosyncratic volatility attenuates negative reactions suggests that markets penalize AI service patents less in firms already perceived as adaptive and innovation-oriented. Managers in such firms can lean into this reputation by explicitly referencing their historical track record of successfully navigating technological uncertainty. Finally, disclosure around AI servitization should be staged. Rather than allowing the patent grant to constitute the first visible signal of strategic intent, firms can acclimate investors through earlier R&D communications, partnership announcements, or conference presentations that gradually articulate the AI service strategy.

For investors the short-term negative reactions we observe need not imply that AI service patents are value-destroying; instead, they may reflect temporary discounting under uncertainty. For investors, this creates a potential opportunity to differentiate between defensive or symbolic patenting—which primarily signals strategic ambiguity—and substantive AI servitization efforts rooted in complementary capabilities. In evaluating such announcements, investors should assess whether patents are supported by verifiable implementation capacity (e.g., human capital, data assets, ecosystem relationships), credible pilot evidence, and clearly articulated commercialization pathways. Where these elements are present, transient negative reactions may offer entry points into firms that are better positioned to capture long-term service-based value.

For policy makers the investor skepticism surrounding AI service patents also has policy relevance. If capital markets systematically discount AI servitization efforts in traditional manufacturing, firms may underinvest in socially valuable innovations that carry high upfront uncertainty and opaque payoffs. Policymakers seeking to support the digital transformation of manufacturing can target these information frictions directly. Potential interventions include supporting industry consortia that develop technical standards, interoperability protocols, and implementation best practices for AI-enabled services; funding demonstration and testbed projects that generate credible "proof points" for the performance and reliability of AI servitization; and designing innovation support instruments that recognize the distinctive integration challenges of embedding AI services into legacy manufacturing systems. Collectively, such measures can reduce informational asymmetries, lower perceived risk, and encourage more efficient levels of investment in AI-driven servitization.

4.3. Limitations and future research directions

The findings of the study have the following limitations. First, the study's reliance on patent data as a proxy for AI-enabled service innovation introduces several potential limitations. Patent data may not always accurately reflect the actual implementation and commercialization of service innovations, as some firms may choose not to patent their innovations or may not file patents for all aspects of their innovations. Additionally, patent data may overlook other forms of service innovation that do not involve patentable inventions, such as process innovations or organizational innovations. Furthermore, the focus on high AI-scoring patents may exclude valuable insights from patents with moderate or low AI scores, as these patents may still represent significant service innovations. Despite the limitations of patents, patents remain the most widely used proxy of innovation efforts.

Second, the study's focus on short-term stock market reactions as the primary measure of firm performance limits the understanding of the long-term impact of AI-enabled service innovation. Short-term stock market reactions may not fully capture the long-term effects of service technology innovations on factors such as revenue growth, profitability, and customer satisfaction. Long-term studies that track the performance of firms over time are needed to fully assess the impact of AI-enabled service innovation on firm performance. Future research should investigate the long-term impact of AI-enabled service innovation on various aspects of firm performance, including revenue growth, profitability, and customer satisfaction. The study identifies idiosyncratic volatility as a key moderating factor. Future research should explore the moderating role of other firm-level characteristics, such as organizational culture, leadership, and resource allocation strategies, on the relationship between service technology patents and stock market reactions.

Third, it is important to note that the contemporaneous endogeneity of the relationship between service technology patents and stock market reactions is not a major concern in this study. This is because the stock market reactions are measured at the time of the patent approval, which is typically several years after the initial investment in the invention. Therefore, it is unlikely that the stock market reactions are directly affected by the patent itself. Instead, the stock market reactions are likely to reflect investor expectations about the future performance of the firm, which may be influenced by the perceived potential of the AI-enabled service innovation.

Fourth, we acknowledge that patent grant announcements may occasionally coincide with other corporate events (earnings releases, partnerships, product launches) that could confound our measured stock reactions. However, several factors mitigate this concern. First, patent grant timing is determined by USPTO examination processes rather than firm strategic scheduling, providing quasi-random timing that reduces endogenous announcement clustering. Second, our firm fixed effects control for firm-specific propensities toward volatile stock behavior or frequent news generation. Third, the matched-pair analysis (Table 5) compares service patents to non-service patents from the same firms, implicitly controlling for firm-level announcement patterns. While we cannot guarantee complete isolation of patent effects, the combination of quasi-random timing, fixed effects, and matched-pair robustness provides reasonable confidence in causal interpretation. Future research with access to comprehensive historical news databases could further address this concern through event contamination filtering.

Finally, the study's findings are based on a sample of US manufacturing firms during a specific time period. The generalizability of the findings to other countries, industries, and time periods may be limited due to variations in market conditions, regulatory environments, and technological advancements. Future research should replicate the study using data from different countries, industries, and time periods to assess the generalizability of the findings and identify any contextual factors that may influence the relationship between AI-enabled service innovation and firm performance.

In conclusion, this study, grounded in transaction cost economics and organizational information processing, challenges the perceived benefits of AI in servitization. Our findings reveal short-term negative stock market reactions to AI-based service patents in manufacturing firms, with this pattern strengthening from the 2010s onward as AI technologies matured. Critically, we measure market responses at patent approval dates, capturing immediate investor sentiment rather than long-term value realization. This short-term focus reflects both a methodological boundary and a substantive finding about capital market perceptions: investors apply skepticism to AI servitization that may not align with eventual operational outcomes.

Whether these negative reactions reflect market efficiency (accurately forecasting implementation difficulties) or market myopia (undervaluing long-term transformation potential) remains an important open question. Future research should track whether negative approval-day reactions reverse as firms demonstrate successful implementation, or whether they prove predictive of actual performance challenges. The decade-wise strengthening suggests markets have become more, not less, skeptical over time—possibly reflecting learned experience with AI hype cycles and the gap between promised and realized benefits. This learned skepticism may protect investors from overvaluation while potentially discouraging worthwhile long-term AI service investments.

CRediT authorship contribution statement

Pankaj C. Patel: Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author has no conflict of interest to declare.

Appendix A. Illustrative examples of AI-Rich service technology patents from manufacturing firms

This appendix provides concrete examples of AI-intensive service technology patents from manufacturing firms to illustrate the theoretical concerns discussed in this study regarding investor reactions to service innovation announcements. These patents exemplify the types of AI-based service technologies that manufacturing firms are developing, and the examples demonstrate how Transaction Cost Economics (TCE) and Organizational Information Processing Theory (OIPT) help explain negative stock market reactions despite the apparent technological sophistication of these innovations.

The examples presented here serve to illustrate the types of AI service innovations captured in our empirical analysis. In our full empirical study, we systematically identify service patents using the Geum et al. (2017) service technology patent classification across ten technology categories, and we measure AI intensity using the Giczy et al. (2022) Artificial Intelligence Patent Dataset (AIPD), which employs validated machine learning classification models across eight AI component technologies. The specific patents illustrated here represent the practical manifestation of our theoretical constructs regarding capability gaps, transaction costs, and information processing challenges that explain negative market reactions to AI service patent approvals.

A.1. Example 1: General electric company

Patent Number: US Patent Application No. 20230252278 (Published August 10, 2023)

Title: “Predictive Maintenance General AI Engine and Method”

Assignee: General Electric

Status: Published application (pending).

A.1.1. Technology description

This patent describes an automated system designed to generate AI-powered predictive maintenance models for industrial machinery. The system processes historical sensor readings and failure logs to build machine learning models capable of predicting equipment failures before they occur. A key innovation is the system's “sensor and machine agnostic” design, meaning it can theoretically work with any type of industrial equipment regardless of manufacturer or sensor configuration. The patent employs multiple AI technology components including machine learning classification models for failure prediction, time-series data analysis capable of processing three-dimensional temporal patterns, both two-dimensional and three-dimensional neural networks, sophisticated feature extraction and selection algorithms, and techniques for augmented failure data generation to address the challenge of rare failure events in training data.

A.1.2. Investor concerns

From a Transaction Cost Economics perspective, this patent raises fundamental questions about GE's organizational capabilities to deliver on this service promise. Investors may question whether GE possesses sufficient data science talent to train and validate failure prediction models across the diverse range of equipment types and operating conditions found in customer facilities. The promise of a “general” AI engine that works across any equipment type is technologically ambitious, potentially signaling overreach beyond GE's core manufacturing expertise. Customer trust represents another critical TCE concern, as industrial customers must be willing to act proactively on AI maintenance recommendations despite the risk of costly false positives that could damage expensive equipment or disrupt production schedules. The system requires extensive labeled failure data for training, raising questions about how GE will acquire sufficient high-quality training examples across various equipment types, operating environments, and failure modes—particularly given that customers may be reluctant to share proprietary operational data.

From an Organizational Information Processing Theory perspective, implementation complexity creates substantial coordination costs. Integrating this system with customers' existing asset management systems, work order systems, and maintenance scheduling processes requires extensive technical coordination across organizational boundaries. Each customer implementation represents a unique integration challenge given heterogeneous IT infrastructures, creating high transaction-specific investments and potential hold-up problems. The patent also exposes monetization uncertainty that concerns investors. It remains unclear whether this represents a standalone service offering with subscription-based pricing, a value-added service bundled with equipment sales, or a cost center designed primarily to reduce warranty claims. Customers may resist paying separately for predictive maintenance services they view as basic equipment support, limiting revenue potential. Finally, the patent reveals competitive moat concerns, as software-based predictive maintenance may face intense competition from pure-play AI service providers like C3.ai or Uptake Technologies that possess superior algorithms, more extensive cross-industry training data, and deeper data science capabilities than a traditional manufacturing firm.

A.2. Example 2: Siemens corporation

Patent Number: US10579928B2 (Issued March 3, 2020)

Title: "Log-Based Predictive Maintenance Using Multiple-Instance Learning"

Assignee: Siemens Corporation (subsequently assigned to Siemens Aktiengesellschaft)

Status: Active patent, expires 2035.

A.2.1. Technology description

This Siemens patent presents a sophisticated data-driven approach for predictive maintenance that mines machine event logs to predict component failures. Unlike sensor-based approaches, this innovation focuses on the rich operational information contained in daily machine logs that are automatically generated during both normal and abnormal equipment operation. The system addresses a fundamental challenge: machine logs are primarily designed for debugging purposes rather than failure prediction, containing heterogeneous data including symbolic sequences, numeric time series, categorical variables, and unstructured text. The patent's key innovation is the application of multiple-instance learning (MIL) algorithms specifically designed to handle uncertain failure labels and extremely imbalanced datasets where failure cases are rare. The system employs L1-regularized support vector machines to build sparse linear models that domain experts can interpret, uses bootstrap feature selection techniques to ensure robust performance despite label noise, and implements sophisticated event log parsing to extract predictive features from thousands of unique event codes and theoretically unlimited distinct message texts. The system groups daily logs into "bags" and applies MIL to learn patterns discriminating normal from abnormal instrument performance, even when the precise timing and nature of failures are uncertain due to imperfect service notification data.

A.2.2. Investor concerns

From an OIPT perspective, this patent reveals substantial domain knowledge requirements that create high coordination costs between data scientists and domain experts. The system requires extensive expertise to interpret event codes specific to each machine type, validate that extracted features genuinely indicate impending failures rather than normal operational variations, tune model parameters for specific equipment and operating contexts, and continuously maintain models as equipment and software are updated. These requirements create organizational information processing demands that may exceed Siemens' current capabilities, particularly when scaling across diverse equipment portfolios. TCE concerns arise from label quality and uncertainty, as the system's reliance on service notifications for training data introduces significant label noise. Inaccurate notification dates, incomplete failure documentation, or failures to log certain types of breakdowns degrade model reliability—a problem especially acute for rare failure events where even small amounts of label noise have outsized impacts on model performance. Customers operating critical equipment may be unwilling to trust predictions trained on noisy historical data.

Generalization challenges represent another investor concern, as models trained on one machine family or operating environment may not transfer well to different equipment types or usage patterns. Customers operating diverse machinery may require dozens or hundreds of custom models, multiplying development costs and ongoing maintenance expenses while reducing economies of scale. From a TCE perspective, customer data access requirements raise substantial concerns about transaction costs and governance. Implementation requires continuous access to customer machine logs, raising concerns about proprietary operational data exposure, cybersecurity vulnerabilities for sensitive manufacturing data, governance costs to establish data sharing agreements and ensure compliance, and potential customer resistance to providing ongoing access to detailed operational data that may reveal production volumes, process parameters, or quality issues they consider confidential.

Model interpretation and trust issues persist despite the patent's emphasis on sparse linear models for interpretability. Customers may still question AI recommendations, particularly for high-stakes decisions involving expensive equipment or safety risks. The "black box" perception of AI may persist even with interpretable models, especially among maintenance personnel accustomed to traditional diagnostic approaches. Finally, the service revenue model raises profitability concerns, as log-based predictive maintenance potentially cannibalizes traditional reactive service revenue. Investors may question whether predictive maintenance increases customer lifetime value sufficiently to offset declining emergency service call revenue, parts sales from catastrophic failures, and expedited shipping fees. The business case requires both demonstrating value to customers and capturing sufficient value through pricing to justify the substantial development and deployment costs.

A.3. Example 3: Deere & company (John Deere)

Patent Number: US10102693B1 (Issued October 16, 2018)

Title: "Predictive Analysis System and Method for Analyzing and Detecting Machine Sensor Failures"

Assignee: Deere & Company

Status: Active patent.

A.3.1. Technology description

This John Deere patent describes a predictive analysis system specifically designed to detect sensor failures and predict component degradation in agricultural, construction, and forestry equipment. The system employs a unique approach that partitions sensor signals into successive time intervals and computes similarity values through comparative analysis between current operational data and stored baseline signals captured during known healthy operation. The patent implements vector space algorithms including cosine similarity functions to quantify the degree of similarity between operational patterns, uses root mean square (RMS) calculations to generate weight ratios that account for signal amplitude differences, and computes a quantified degree of failure (DOF) metric that serves as an early warning indicator. The system architecture includes a remote central processing system that receives sensor data from a vehicle electronics unit via wireless infrastructure, processes the data using predictive diagnostic algorithms stored in remote data storage, and returns maintenance recommendations and failure predictions to operators through a user interface. The patent demonstrates particular sophistication in addressing the challenge of distinguishing genuine failure signals from normal operational variations and measurement noise.

A.3.2. Investor concerns

From a TCE perspective, this patent exposes significant customer dependency and lock-in concerns that may worry investors about long-term competitive dynamics. The system requires continuous telematics connectivity between equipment and John Deere's remote processing system, creating ongoing service dependencies. Customers operating in remote agricultural or forestry locations with poor cellular connectivity may experience degraded service quality or be unable to use the system effectively. This creates potential customer dissatisfaction and limits the addressable market. The requirement for baseline data captured during manufacturing or initial operation creates switching costs, as customers with extensive baseline data libraries would face substantial transition costs if switching to competitive equipment, but this dependency may also generate customer resentment and regulatory scrutiny regarding data portability and interoperability.

From an OIPT perspective, the multi-sensor integration complexity creates substantial information processing demands. Modern agricultural equipment employs dozens or hundreds of sensors monitoring hydraulic pressure, engine temperature, implement position, fuel injection timing, transmission status, and numerous other parameters. Interpreting complex sensor interactions across diverse operating conditions (soil types, crop types, terrain, weather) requires sophisticated analysis capabilities that may exceed John Deere's traditional engineering expertise. The patent's reliance on baseline signals captured during steady-state operation raises practical implementation challenges, as establishing what constitutes “steady state” varies dramatically across equipment types and applications. Harvesting equipment, for example, experiences wildly different loading conditions depending on crop density, moisture content, and ground speed, making baseline establishment problematic.

Competitive positioning concerns arise because sensor failure detection, while valuable, represents table stakes functionality that customers increasingly expect as standard equipment capability rather than a premium service offering. Competitors including CNH Industrial, AGCO, and increasingly, pure-play agricultural technology companies like Trimble and Raven Industries, offer similar telematics and predictive maintenance capabilities. Investors may question whether this patent provides sustainable competitive advantage or merely prevents John Deere from falling behind competitors. The business model uncertainty compounds these concerns, as the patent does not clearly indicate whether sensor failure prediction is monetized through subscription services, bundled with equipment purchases, or offered free as a basic customer support function. If offered as a subscription service, customer willingness to pay remains uncertain, particularly for smaller operators facing tight margins. If bundled with equipment, it may not create differentiated value sufficient to justify premium equipment pricing.

From a TCE perspective, liability and trust issues loom large in agricultural applications where equipment failures during critical windows (planting, spraying, harvesting) can have catastrophic financial consequences for farmers. If the system generates false positives that cause unnecessary downtime during harvest season, or false negatives that fail to predict failures, John Deere could face substantial liability exposure and customer trust erosion. The patent's emphasis on quantified degree of failure metrics suggests probabilistic rather than deterministic predictions, creating inherent uncertainty that farmers accustomed to mechanical reliability may resist. Finally, the data privacy and ownership concerns common across agricultural telematics worry investors about regulatory risk and customer backlash. Farmers increasingly voice concerns about equipment manufacturers collecting detailed operational data that reveals proprietary farming practices, field productivity, and financial performance, raising questions about data ownership, monetization of farmer data for agricultural insights, regulatory intervention similar to European “right to repair” legislation, and potential antitrust concerns if John Deere leverages operational data to gain advantages in adjacent markets like precision agriculture services or agricultural lending.

A.4. Cross-cutting theoretical insights

Across these three patents from General Electric, Siemens, and Deere & Company, consistent patterns emerge that illuminate why investors react negatively to AI service patent announcements from manufacturing firms despite apparent technological sophistication. From a Transaction Cost Economics perspective, all three patents demonstrate high asset specificity in implementation, as each requires customer-specific data collection, model training, and systems integration creating switching costs and potential hold-up problems. All three face uncertainty in performance that depends on training data quality, the representativeness of historical failures, customer adoption rates, and the evolving capabilities of AI technologies. Each patent reveals substantial governance costs for managing vendor relationships, establishing data security protocols, ensuring regulatory compliance with data privacy regulations like GDPR or sector-specific requirements, and negotiating service level agreements. Finally, all three expose opportunism risks as customers fear vendor lock-in through proprietary data dependencies and potential misuse of operational data, while vendors fear customer free-riding on intellectual property and customer churn after knowledge transfer.

From an Organizational Information Processing Theory perspective, these patents reveal information processing demands that likely exceed manufacturing firms' traditional organizational capabilities. Each requires integration across organizational boundaries, coordinating between equipment manufacturers, IT service providers, maintenance service organizations, and customer operations teams. All three depend on requisite variety in expertise, needing combinations of domain engineering knowledge, data science and machine learning capabilities, software development and cloud infrastructure management, and customer success and change management skills. The patents demonstrate equivocality in problem definition, as customer “requirements” for predictive maintenance vary dramatically by industry, application, risk tolerance, and existing maintenance practices, making standardized solutions difficult. Implementation requires substantial organizational learning and adaptation to develop new capabilities in data science and AI, build new service delivery organizations and business models, and establish new customer relationships based on ongoing service rather than episodic equipment sales.

These theoretical insights help explain the market reaction paradox at the heart of this study. While these patents demonstrate genuine technological innovation and address real customer pain points around equipment downtime and maintenance costs, they simultaneously signal substantial organizational challenges and uncertain monetization paths that investors rationally discount. The negative stock market reactions captured in our empirical analysis reflect investor skepticism not about the technology per se, but about manufacturing firms' organizational readiness to successfully commercialize AI-intensive services that require capabilities, business models, and customer relationships fundamentally different from their traditional product-centric operations.

Data availability

The authors do not have permission to share data.

References

Abood, A., Feltenberger, D., 2018. Automated patent landscaping. Artif. Intell. Law 26 (2), 103–125.
Abou-Foul, M., Ruiz-Alba, J.L., López-Tenorio, P.J., 2023. The impact of artificial intelligence capabilities on servitization: the moderating role of absorptive capacity- a dynamic capabilities perspective. J. Bus. Res. 157, 113609.
Agarwal, G.K., Simonsson, J., Magnusson, M., Hald, K.S., Johanson, A., 2022. Value-capture in digital servitization. J. Manuf. Technol. Manag. 33 (5), 986–1004.
Akter, S., Hossain, M.A., Sajib, S., Sultana, S., Rahman, M., Vrontis, D., McCarthy, G., 2023. A framework for AI-powered service innovation capability: review and agenda for future research. Technovation 125, 102768.
Alexiev, A.S., Janssen, M., den Hertog, P., 2018. The moderating role of tangibility in synchronous innovation in services. J. Prod. Innov. Manag. 35 (5), 682–700.
Almeida, H., Campello, M., 2007. Financial constraints, asset tangibility, and corporate investment. Rev. Financ. Stud. 20 (5), 1429–1460.
Atasoy, Ö., Trudel, R., Noseworthy, T.J., Kaufmann, P.J., 2022. Tangibility bias in investment risk judgments. Organ. Behav. Hum. Decis. Process. 171, 104150.
Baines, T., Ziaee Bigdeli, A., Bustinza, O.F., Shi, V.G., Baldwin, J., Ridgway, K., 2017. Servitization: revisiting the state-of-the-art and research priorities. Int. J. Oper. Prod. Manag. 37 (2), 256–278.
Baines, T.S., Lightfoot, H.W., Benedettini, O., Kay, J.M., 2009. The servitization of manufacturing: a review of literature and reflection on future challenges. J. Manuf. Technol. Manag. 20 (5), 547–567.
Beuren, F.H., Ferreira, M.G.G., Miguel, P.A.C., 2013. Product-service systems: a literature review on integrated products and services. J. Clean. Prod. 47, 222–231.
Bonacich, P., 1987. Power and centrality: a family of measures. Am. J. Sociol. 92 (5), 1170–1182.
Borgatti, S.P., Li, X., 2009. On social network analysis in a supply chain context. J. Supply Chain Manag. 45 (2), 5–22.
Brax, S.A., Calabrese, A., Levialdi Ghiron, N., Tiburzi, L., Grönroos, C., 2021. Explaining the servitization paradox: a configurational theory and a performance measurement framework. Int. J. Oper. Prod. Manag. 41 (5), 517–546.
Burt, R.S., 1992. Structural holes. In: In structural holes. Structural Holes.
Burton, R.M., Obel, B., 2018. The science of organizational design: fit between structure and coordination. J. Organ. Des. 7 (1), 1–13.
Bustinza, O.F., Vendrell-Herrero, F., Gomes, E., Lafuente, E., Opazo-Basáez, M., Rabetino, R., Vaillant, Y., 2018. Product-service innovation and performance: unveiling the complexities. Int. J. Bus. Environ. 10 (2), 95–111.
Carhart, M.M., 1997. On persistence in mutual fund performance. J. Financ. 52 (1), 57–82.
Christensen, J., 2018. A Behavioral Theory of Human Capital Integration. Copenhagen Business School [Phd].
Daft, R.L., Lengel, R.H., 1986. Organizational information requirements, media richness and structural design. Manag. Sci. 32 (5), 554–571.
Daft, R.L., Macintosh, N.B., 1981. A tentative exploration into the amount and equivocality of information processing in organizational work units. Adm. Sci. Q. 207–224.
Dai, Y., 2021. Three Essays on Stock Returns and Idiosyncratic Risk. University of Exeter (United Kingdom).
Dalenogare, L.S., Le Dain, M.-A., Ayala, N.F., Pezzotta, G., Frank, A.G., 2023. Building digital servitization ecosystems: an analysis of inter-firm collaboration types and social exchange mechanisms among actors. Technovation 124, 102756.
Daniel, K., Titman, S., 2006. Market reactions to tangible and intangible information. J. Financ. 61 (4), 1605–1643.
Dmitrijeva, J., Schroeder, A., Bigdeli, A.Z., Baines, T., 2022. Paradoxes in servitization: a processual perspective. Ind. Mark. Manag. 101, 141–152.
Faramarzi, A., Worm, S., Ulaga, W., 2023. Service strategy's effect on firm performance: a meta-analysis of the servitization literature. J. Acad. Mark. Sci. 1–27.
Flynn, B.B., Flynn, E.J., 1999. Information-processing alternatives for coping with manufacturing environment complexity. Decis. Sci. 30 (4), 1021–1052.
Freeman, L.C., Roeder, D., Mulholland, R.R., 1979. Centrality in social networks: II. Experimental results. Social networks 2 (2), 119–141.
Galbraith, J., 1973. Designing Complex Organizations. Reading, Mass.
Galbraith, J.R., 1977. Organization design. Reading, Ma.
Gebauer, H., Paiola, M., Saccani, N., Rapaccini, M., 2021. Digital servitization: Crossing the perspectives of digitization and servitization. In: In, vol. 93. Elsevier, pp. 382–388.
Geum, Y., Kim, M.-S., Lee, S., 2017. Service technology: definition and characteristics based on a patent database. Serv. Sci. 9 (2), 147–166.
Giczy, A.V., Pairolero, N.A., Toole, A.A., 2022. Identifying artificial intelligence (AI) invention: A novel AI patent dataset. J. Technol. Transf. 47 (2), 476–505.
Grant, R.M., 2006. The Knowledge-Based View of the Firm. A strategy overview and competitive strategy, The Oxford handbook of strategy, pp. 203–230.
Haußmann, C., Dwivedi, Y.K., Venkitachalam, K., Williams, M.D., 2012. A summary and review of Galbraith's organizational information processing theory. Information Systems Theory: Explaining and Predicting Our Digital Society 2, 71–93.
Jacobs, B.W., Kraude, R., Narayanan, S., 2016. Operational productivity, corporate social performance, financial performance, and risk in manufacturing firms. Prod. Oper. Manag. 25 (12), 2065–2085.
Jancenelle, V.E., 2021. Tangible− intangible resource composition and firm success. Technovation 108, 102337.
Johnson, M., Roehrich, J.K., Chakkol, M., Davies, A., 2021. Reconciling and reconceptualising servitization research: drawing on modularity, platforms, ecosystems, risk and governance to develop mid-range theory. Int. J. Oper. Prod. Manag. 41 (5), 465–493.
Jorzik, P., Klein, S.P., Kanbach, D.K., Kraus, S., 2024. AI-driven business model innovation: a systematic review and research agenda. J. Bus. Res. 182, 114764.
Joseph, J., Gaba, V., 2020. Organizational structure, information processing, and decision-making: a retrospective and road map for research. Acad. Manag. Ann. 14 (1), 267–302.
Josephson, B.W., Johnson, J.L., Mariadoss, B.J., Cullen, J., 2016. Service transition strategies in manufacturing: implications for firm risk. J. Serv. Res. 19 (2), 142–157.
Kamp, B., Parry, G., 2017. Servitization and advanced business services as levers for competitiveness. Ind. Mark. Manag. 60, 11–16.
Khanra, S., Dhir, A., Parida, V., Kohtamäki, M., 2021. Servitization research: a review and bibliometric analysis of past achievements and future promises. J. Bus. Res. 131, 151–166.
Kharlamov, A.A., Parry, G., 2021. The impact of servitization and digitization on productivity and profitability of the firm: a systematic approach. Prod. Plan. Control 32 (3), 185–197.
Kinkel, S., Baumgartner, M., Cherubini, E., 2022. Prerequisites for the adoption of AI technologies in manufacturing–evidence from a worldwide sample of manufacturing companies. Technovation 110, 102375.
Kmetz, J.L., 2018. The Information Processing Theory of Organization: Managing Technology Accession in Complex Systems. Routledge.
Kogan, L., Papanikolaou, D., Seru, A., Stoffman, N., 2017. Technological innovation, resource allocation, and growth. Q. J. Econ. 132 (2), 665–712.
Kohtamäki, M., Einola, S., Rabetino, R., 2020. Exploring servitization through the paradox lens: coping practices in servitization. Int. J. Prod. Econ. 226, 107619.
Kohtamäki, M., Henneberg, S.C., Martinez, V., Kimita, K., Gebauer, H., 2019. A configurational approach to servitization: review and research directions. Serv. Sci. 11 (3), 213–240.
Kohtamäki, M., Parida, V., Oghazi, P., Gebauer, H., Baines, T., 2019b. Digital servitization business models in ecosystems: a theory of the firm. J. Bus. Res. 104, 380–392.
Kroh, J., Luetjen, H., Globocnik, D., Schultz, C., 2018. Use and efficacy of information technology in innovation processes: the specific role of servitization. J. Prod. Innov. Manag. 35 (5), 720–741.
Laroche, M., Bergeron, J., Goutaland, C., 2003. How intangibility affects perceived risk: the moderating role of knowledge and involvement. J. Serv. Mark. 17 (2), 122–140.
Lexutt, E., 2019. *Success and Failure of Servitization* Dissertation, FernUniversität in Hagen, 2019.
Lim, D., 2018. AI & IP: innovation & creativity in an age of accelerated change. Akron L. Rev. 52, 813.
Luiz Corrêa, H., Ellram, L.M., José Scavarda, A., Cooper, M.C., 2007. An operations management view of the services and goods offering mix. Int. J. Oper. Prod. Manag. 27 (5), 444–463.
Marinotti, J., 2020. Tangibility as technology. Ga. St. UL Rev. 37, 671.
Mazzucato, M., Tancioni, M., 2008. Innovation and idiosyncratic risk: an industry-and firm-level analysis. Ind. Corp. Chang. 17 (4), 779–811.
Miller, K.D., Chen, W.-R., 2003. Risk and firms' costs. Strateg. Organ. 1 (4), 355–382.
Neely, A., 2008. Exploring the financial consequences of the servitization of manufacturing. Oper. Manag. Res. 1, 103–118.
Ocasio, W., 2001. How Do Organizations Think. Computation and interpretation, Organizational cognition, pp. 39–60.
Orton, J.D., Weick, K.E., 1990. Loosely coupled systems: a reconceptualization. Acad. Manag. Rev. 15 (2), 203–223.
Paschou, T., Rapaccini, M., Adrodegari, F., Saccani, N., 2020. Digital servitization in manufacturing: a systematic literature review and research agenda. Ind. Mark. Manag. 89, 278–292.
Patel, P.C., Pearce, J.A., Guedes, M.J., 2019. The survival benefits of service intensity for new manufacturing ventures: a resource-advantage theory perspective. J. Serv. Res. 22 (4), 352–370.
Rabetino, R., Harmsen, W., Kohtamäki, M., Sihvonen, J., 2018. Structuring servitization-related research. Int. J. Oper. Prod. Manag. 38 (2), 350–371.
Rabetino, R., Kohtamäki, M., Brax, S.A., Sihvonen, J., 2021. The tribes in the field of servitization: discovering latent streams across 30 years of research. Ind. Mark. Manag. 95, 70–84.
Raddats, C., Kowalkowski, C., Benedettini, O., Burton, J., Gebauer, H., 2019. Servitization: a contemporary thematic review of four major research streams. Ind. Mark. Manag. 83, 207–223.
Raddats, C., Naik, P., Bigdeli, A.Z., 2022. Creating value in servitization through digital service innovations. Ind. Mark. Manag. 104, 1–13.
Rojas-Córdova, C., Williamson, A.J., Pertuze, J.A., Calvo, G., 2023. Why one strategy does not fit all: a systematic review on exploration–exploitation in different organizational archetypes. Rev. Manag. Sci. 17 (7), 2251–2295.
Ross, J.M., Fisch, J.H., Varga, E., 2018. Unlocking the value of real options: how firm-specific learning conditions affect R&D investments under uncertainty. Strateg. Entrep. J. 12 (3), 335–353.
Ruiz-Martín, A., Díaz-Garrido, E., 2021. A review of servitization theoretical foundations. Journal of Industrial Engineering and Management (JIEM) 14 (3), 496–519.
Rüttenauer, T., Ludwig, V., 2020. Fixed effects individual slopes: accounting and testing for heterogeneous effects in panel data or other multilevel models. Sociol. Methods Res. 0049124120926211.
Schaefers, T., Leban, M., Vogt, F., 2022. On-demand features: consumer reactions to tangibility and pricing structure. J. Bus. Res. 139, 751–761.
Sjödin, D., Parida, V., Kohtamäki, M., 2019. Relational governance strategies for advanced service provision: multiple paths to superior financial performance in servitization. J. Bus. Res. 101, 906–915.
Sjödin, D., Parida, V., Kohtamäki, M., 2023. Artificial intelligence enabling circular business model innovation in digital servitization: conceptualizing dynamic

capabilities, AI capacities, business models and effects. Technol. Forecast. Soc. Chang. 197, 122903.
Stock, G.N., Tatikonda, M.V., 2008. The joint influence of technology uncertainty and interorganizational interaction on external technology integration success. J. Oper. Manag. 26 (1), 65–80.
Teece, D.J., Pisano, G., Shuen, A., 1997. Dynamic capabilities and strategic management. Strateg. Manag. J. 18 (7), 509–533.
Ulaga, W., Kowalkowski, C., 2022. Servitization: a state-of-the-art overview and future directions. The Palgrave Handbook of Service Management 169–200.
Vaittinen, E., Nenonen, S., 2018. Antecedents of consumer acceptance of a manufacturer's goods-related service. Serv. Mark. Q. 39 (4), 310–329.
Vandermerwe, S., Rada, J., 1988. Servitization of business: adding value by adding services. Eur. Manag. J. 6 (4), 314–324.
Vargo, S.L., Lusch, R.F., 2008. Service-dominant logic: continuing the evolution. J. Acad. Mark. Sci. 36, 1–10.
Vieira Da Cunha, J., Kamoche, K., Pina, E., Cunha, M., 2003. Organizational improvisation: a contextual approach. Int. Rev. Sociol. 13 (3), 567–589.
Villalonga, B., 2004. Intangible resources, Tobin'sq, and sustainability of performance differences. J. Econ. Behav. Organ. 54 (2), 205–230.
Wan, J., Li, X., Dai, H.-N., Kusiak, A., Martinez-Garcia, M., Li, D., 2020. Artificial-intelligence-driven customized manufacturing factory: key technologies, applications, and challenges. Proc. IEEE 109 (4), 377–398.
Wang, W., Lai, K.-H., Shou, Y., 2018. The impact of servitization on firm performance: a meta-analysis. Int. J. Oper. Prod. Manag. 38 (7), 1562–1588.
Wickens, C.D., Carswell, C.M., 2021. Information processing. Handbook of human factors and ergonomics 114–158.
Williamson, O., 1985. The Economic Institutions of Capitalism. Free Press, New York.
Xing, Y., Liu, Y., Davies, P., 2023. Servitization innovation: a systematic review, integrative framework, and future research directions. Technovation 122, 102641.
Xu, Y., Malkiel, B.G., 2003. Investigating the behavior of idiosyncratic volatility. J. Bus. 76 (4), 613–645.

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